

S3

Benvingut al núvol. En aquest Sprint, faràs el salt de treballar en local (MySQL) a treballar amb **Google BigQuery**, un Data Warehouse serverless dissenyat per a l'anàlisi de dades a gran escala.

En l'entorn empresarial actual, l'analista no acostuma a crear bases de dades des de zero, sinó que ha de connectar-se a repositoris existents (Data Lakes), comprendre els costos del núvol i preparar les dades brutes per a l'anàlisi (ELT).

Lliurament

Revisa el material a entregar en cada exercici.

- **Codi i Infraestructura (.sql)** Els arxius .sql amb tots els scripts generats.
- **Evidència d'Execució i Rendiment (PDF)** Un document PDF únic que contingui les captures de pantalla de la Consola de BigQuery per a cada exercici.
 - **Requisit Crític:** Per validar que la teva optimització funciona, la captura ha de mostrar clarament:
 - El **codi SQL** a l'editor.
 - El missatge de validació (check verd ) o la pestanya Job Information on es vegin els Bytes processats (Bytes processed).

Important

Recorda revisar les [millors pràctiques](#) abans de fer el lliurament.

- **Control de Costos:** A BigQuery, cada consulta costa diners. Abans d'executar res, fixa't sempre en el missatge del validador (Dry Run) a dalt a la dreta.
- **Ubicació de les Dades:** Assegura't que el teu Dataset estigui a la regió EU (Multiple regions) o les connexions fallaran.

Limitacions del Sandbox

La zona de proves de BigQuery (Sandbox) està subjecta als següents límits:

- S'apliquen tots els límits i quotes habituals de BigQuery.
- Tens 10 GB d'emmagatzematge actiu i 1 TB de dades de consulta processades cada mes sense cap cost.
-  **Caducitat:** Totes les taules, vistes i particions caduquen de forma automàtica després de **60 dies**.
- No admet funcions avançades com: Transmissió de dades (Streaming), DML (UPDATE / DELETE massius) o el Servei de transferència de dades.

Nivell 1: Entorn i Ingesta Híbrida (Code-First)

Escenari:

Acabeu d'arribar a l'empresa. L'equip d'Enginyeria de Dades ha deixat els fitxers històrics al Data Lake corporatiu.

La teva primera missió és dissenyar i crear l'entorn de treball. El CTO exigeix professionalitat: no abocarem dades sense ordre, sinó que implementarem una Arquitectura Medallion.

Exercici 1: Arquitectura de Dades (Lògica vs. Física)

Abans de tocar la consola, has d'entendre la diferència entre allò que dissenyem i allò que construïm:

- **Arquitectura Lògica (Capes de Dades):** És el disseny conceptual. Organitza les dades segons el seu nivell de qualitat (Bronze, Silver, Gold).
- **Implementació Física (Datasets):** És el recurs real a Google Cloud. A BigQuery, utilitzarem Datasets (contenidors) per materialitzar físicament aquestes capes lògiques.



Tasques:

Implementareu físicament l'arquitectura lògica creant tres Datasets dins d'un mateix projecte per centralitzar la facturació i els permisos.



Crea un **Nou Projecte** a Google Cloud anomenat:

```
sprint3-analytics-[el-teu-nom] .
```

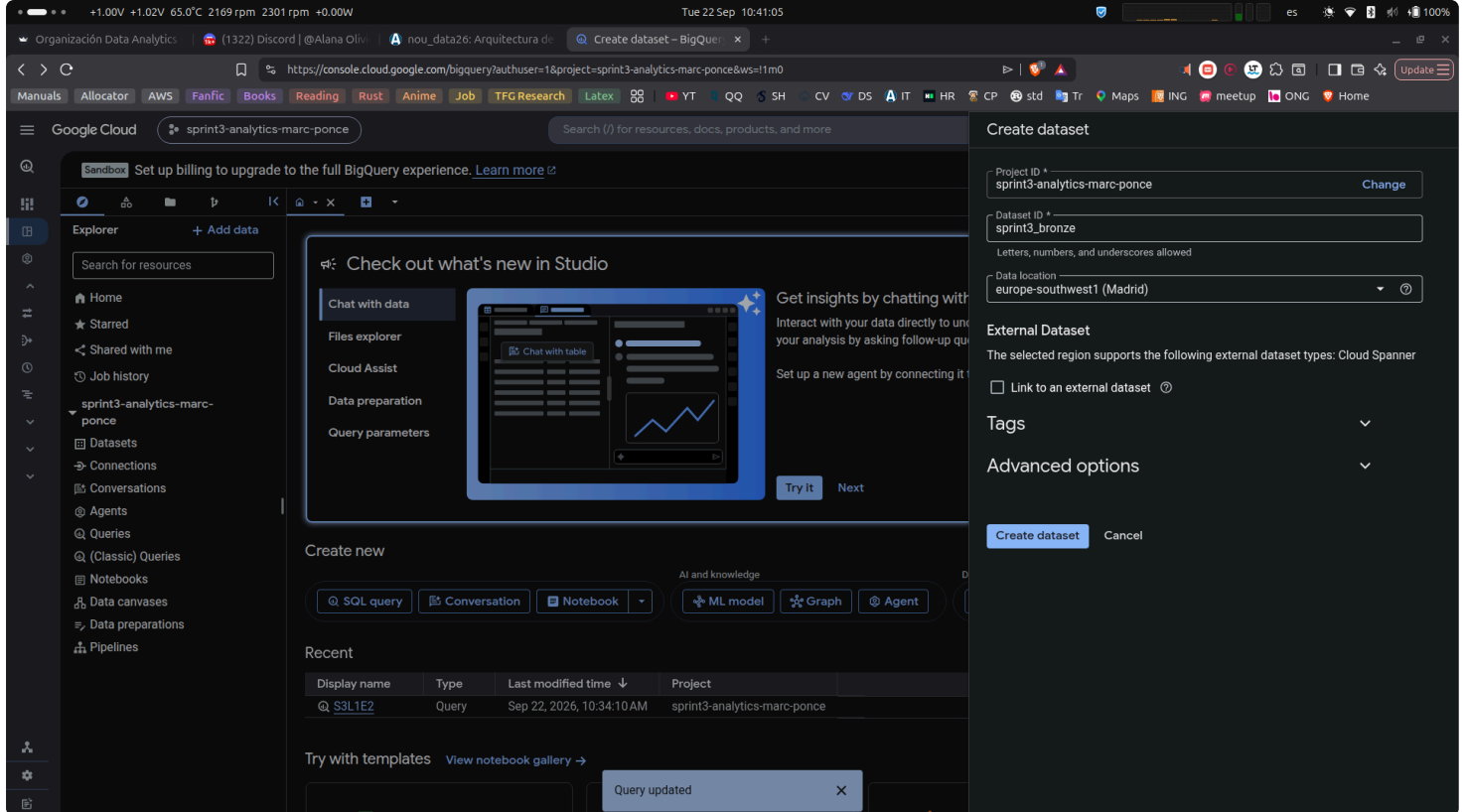
Ho pots crear per la UI, però si ho fas per Consola (gcloud CLI) té un plus.

```
gcloud projects create sprint3-analytics-marc-ponce  
gcloud config set project sprint3-analytics-marc-ponce
```

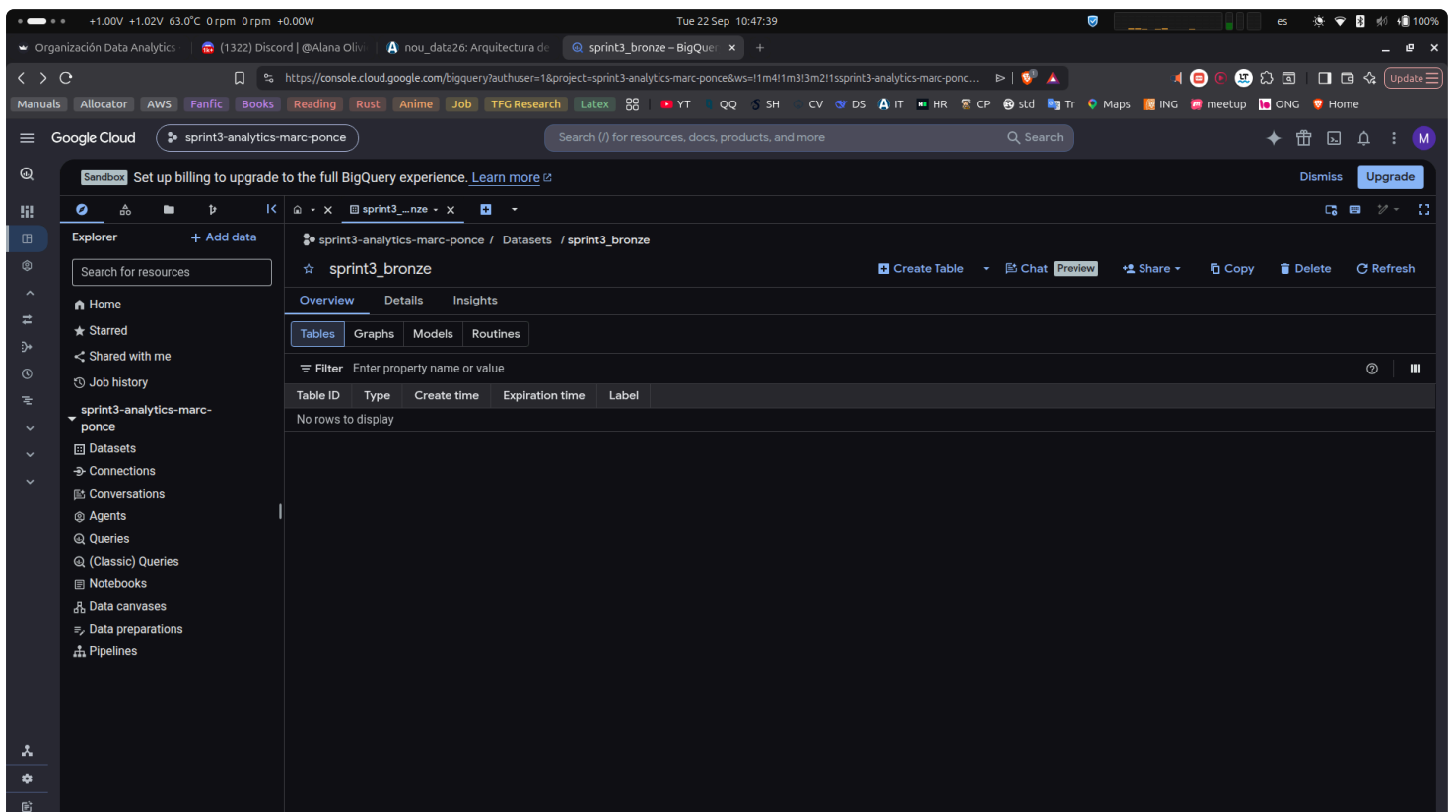
Per demostrar el teu domini total de la plataforma, crearàs cadascun dels datasets utilitzant un mètode diferent:

- **Dataset Físic** `sprint3_bronze` → Capa Lògica Bronze (Dades Crues)
 - Funció: Dades crues tal com arriben de la font.
 - Mètode: Utilitza la **Interfície Gràfica (UI)** de BigQuery (fent clics).

» Click the three points icon in `Datasets` on the left panel, then `Create dataset` . Input the name of the `SCHEMA` , and its location to an EU server: `europa-southwest1` (Madrid).



» The SCHEMA was created successfully:



- **Dataset Físic** sprint3_silver → Capa Lògica Silver (Dades Netes)
 - Funció: Dades netes, tipades i de-duplicades.
 - Mètode: Utilitza codi SQL (CREATE SCHEMA).

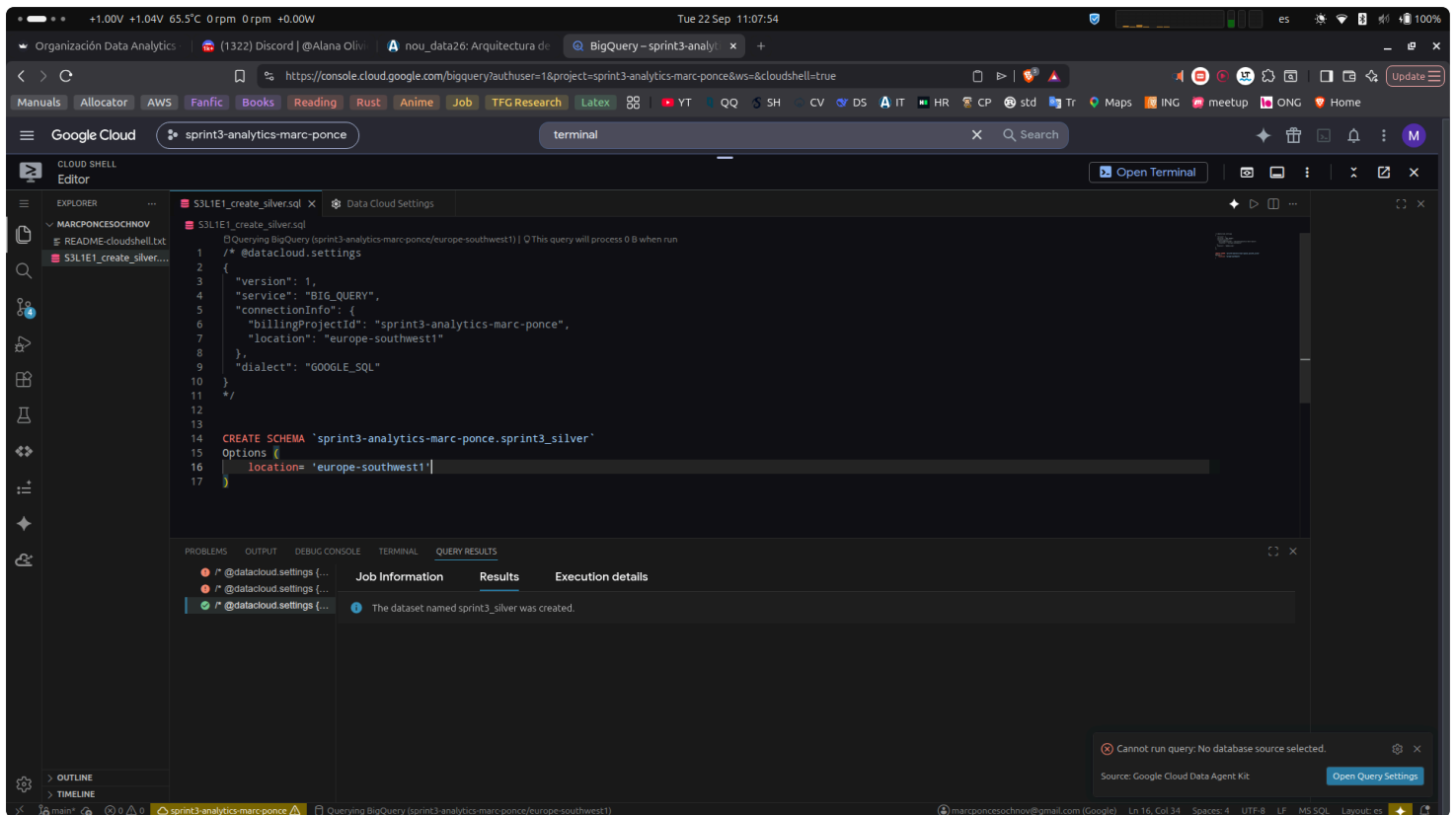
```
/* @datacloud.settings
{
  "version": 1,
  "service": "BIG_QUERY",
  "connectionInfo": {
```

```

    "billingProjectId": "sprint3-analytics-marc-ponce",
    "location": "europe-southwest1"
  },
  "dialect": "GOOGLE_SQL"
}
*/

CREATE SCHEMA `sprint3-analytics-marc-ponce.sprint3_silver`
Options (
  location= 'europe-southwest1'
)

```



- **Dataset Físic** sprint3_gold → Capa Lògica Gold (Dades de Negoci)
 - Funció: Dades agregades llestes per a informes i panells de control (dashboards).
 - Mètode: Utilitza Cloud Shell (Línia d'ordres bq).

```

bq --location=europe-southwest1 \
mk --dataset sprint3-analytics-marc-ponce:sprint3_gold

```

```
projects/p/locations/1/transferConfigs/c
bq update --transfer_config --target-dataset=dataset
--refresh window days=5 --update-credentials
projects/p/locations/1/transferConfigs/c
bq update --reservation --location=US --project_id=my-project
--sl_reservation_size=25
bq update --capacity_commitment --location=US --project_id=my-project
--plan=MONTHLY --renewal-plan=FLEX commitment_id
bq update --capacity_commitment --location=US --project_id=my-project
--split --slots=500 Commitment_id
bq update --capacity_commitment --location=US --project_id=my-project
--merge commitment_id commitment_id2
bq update --reservation_assignment
--destination_reservation_id=proj:US.new_reservation
proj:US.old_reservation.assignment_id
bq update --reservation_assignment
--destination_reservation_id=proj:US.new_reservation
proj:US.old_reservation.assignment_id
--precedence_id
--condition="(Expression: 'job.label'=='prod')'"
bq update --connection_credential={"username":"u", "password":"p"}
--location=US --project_id=my-project existing_connection
bq update --row_access_policy --policy_id=existing_policy
--target-table='existing-dataset.existing-table'
--grantees='user:user@google.com,group:group@google.com'
--filter_predicate='Region=US'

Return the version of bq.

wait

Poll job_id until either (1) the job is DONE or (2) the specified number of seconds have elapsed. Waits forever if unspecified. If no job_id is specified, and there is only one running job, we poll that job.

Examples:
bq wait # Waits forever for the currently running job.
bq wait job_id # Waits forever
bq wait job_id 100 # Waits 100 seconds
bq wait job_id 0 # Polls if a job is done, then returns immediately.
# These may exit with a non-zero status code to indicate "failure":
bq wait --fail_on_error job_id # Succeeds if job succeeds.
bq wait --fail_on_error job_id 100 # Succeeds if job succeeds in 100 sec.

Arguments:
job_id: Job ID to wait on.
secs: Number of seconds to wait (must be >= 0).

Run 'bq.py --help' to get help for global flags.
Run 'bq.py help <command>' to get help for <command>.
marcponcesochnov@cloudshell: (sprint3-analytics-marc-ponce) $ bq --location=europe-southwest1 mk --dataset sprint3-analytics-marc-ponce:gold
Dataset name: sprint3-analytics-marc-ponce:gold is invalid, must be letters (uppercase or lowercase), numbers, and underscores up to 1024 characters.
marcponcesochnov@cloudshell: (sprint3-analytics-marc-ponce) $ bq --location=europe-southwest1 mk --dataset sprint3-analytics-marc-ponce:sprint3_gold
Dataset "sprint3-analytics-marc-ponce:sprint3_gold" successfully created.
marcponcesochnov@cloudshell: (sprint3-analytics-marc-ponce) $
```

» In this last screenshot you can confirm that the datasets have been created successfully:

Organización Data Analytics | (1322) Discord | @Alana Oliv | nou_data26: Arquitectura de | BigQuery – sprint3-analyt...

https://console.cloud.google.com/bigquery/authuser=1&project=sprint3-analytics-marc-ponce&ws=1m6l1m5l16m3l1m11sprint3-analytics-...

Manuals | Allocator | AWS | Fanfic | Books | Reading | Rust | Anime | Job | TFG Research | Latex | YT | QQ | SH | CV | DS | IT | HR | CP | std | Tr | Maps | ING | meetup | ONG | Home

Google Cloud | sprint3-analytics-marc-ponce | terminal

Open Editor

Explorer + Add data

Search for resources

Home

Starred

Shared with me

Job history

sprint3-analytics-marc-ponce

Datasets

Connections

Conversations

Agents

Queries

(Classic) Queries

Notebooks

Data canvases

Data preparations

Pipelines

sprint3-analytics-marc-ponce / Datasets

+ Create dataset Refresh

Clicking a link in the list will replace the current tab. To keep the current tab open, double-click the *tab name*, or use Ctrl+click to open the link in a new tab.

Dismiss

Filter Enter property name or value

Dataset ID	Source	Type	Location	Create time	Label	
sprint3_bronze	BigQuery	Default	europe-southwest1		None	☆
sprint3_gold	BigQuery	Default	europe-southwest1		None	☆
sprint3_silver	BigQuery	Default	europe-southwest1		None	☆

Rows per page: 50 1 – 3 of 3

Tingueu en compte la ubicació de les dades originals per decidir la ubicació física dels tres Datasets.

📌 **Nota:** Encara que siguin capes lògiques diferents, físicament han de viure a la mateixa regió per poder "parlar" entre elles sense costos de transferència.

Exercici 2: Ingesta en Capa Bronze (Connexió DDL)

Un analista professional automatitza les tasques. Ara que tens la teva arquitectura a punt, poblarem la primera capa (**Bronze**) sense moure les dades originals, creant **Taules Externes** que apuntin al Data Lake.

🌱 Tasca:

Escriu i executa les sentències `CREATE EXTERNAL TABLE` per connectar els següents fitxers al *dataset* `sprint3_bronze` . Para molta atenció a les "Notes Tècniques", ja que no tots els arxius tenen el mateix format.

Taula	Domini	Ruta al Núvol (URI)	Notes Tècniques
<code>transactions_raw</code>	ERP	transactions.csv	Delimitador: <code>;</code>
<code>companies_raw</code>	ERP	companies.csv	Capçalera: Ignora la 1a fila. Defineix l'esquema manualment perquè tot és text.
<code>american_users_raw</code>	CRM	american_users.csv	Estàndard.
<code>european_users_raw</code>	CRM	european_users.csv	Estàndard.
<code>credit_cards_raw</code>	CRM	credit_cards.csv	Estàndard.

👉 Entrega:

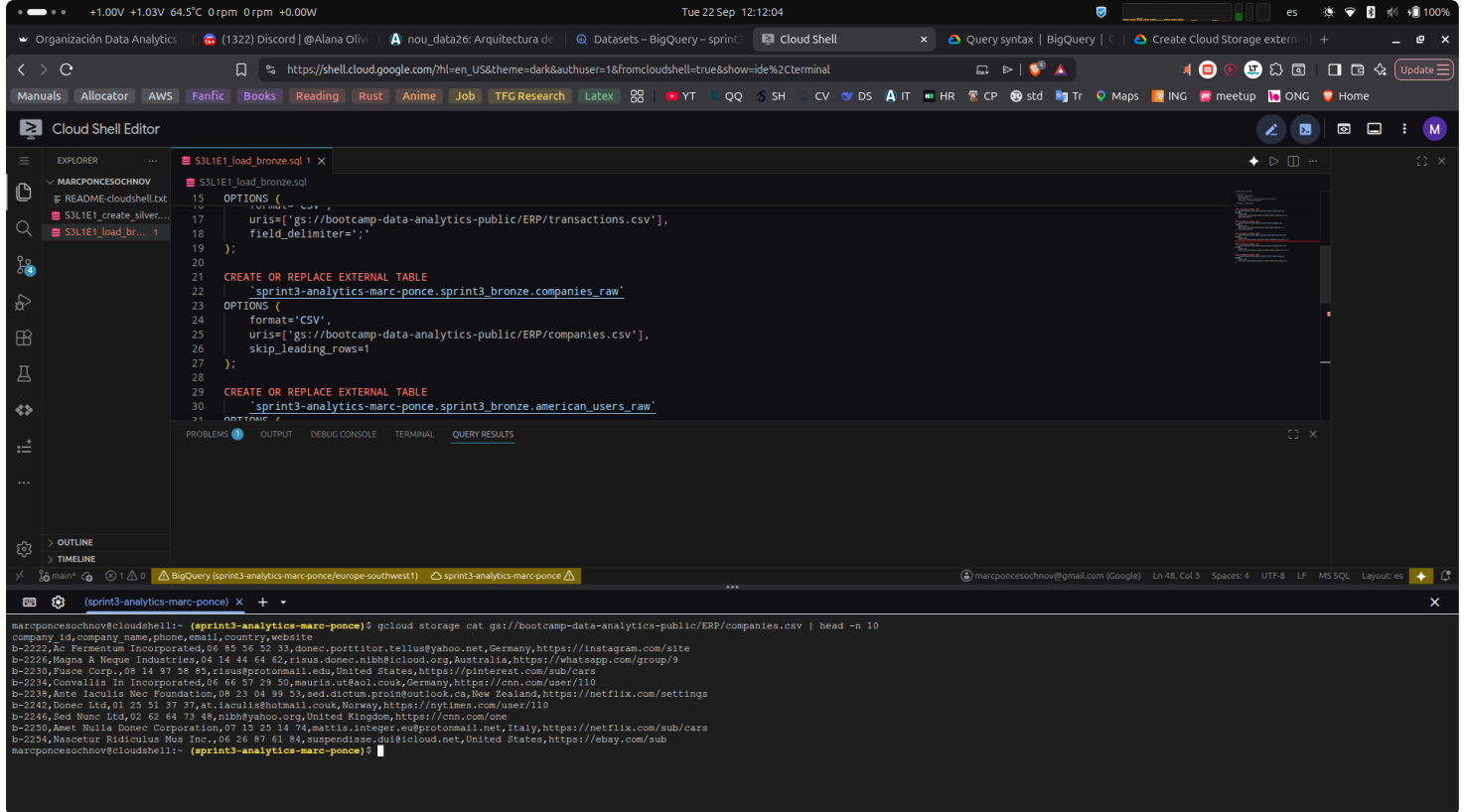
El codi SQL utilitzat per a `transactions_raw` i una captura de pantalla amb les 5 taules creades.

» To understand the structure of `companies.csv` , we `cat | head` the file in the terminal:

```
gcloud storage cat gs://bootcamp-data-analytics-public/ERP/companies.csv | head -n 10
```

» Here is the resulting table:

company_id	company_name	phone	email	country	website
b-2222	Ac Fermentum Incorporated	06 85 56 52 33	donec.porttitor.tellus@yahoo.net	Germany	https://insta
b-2226	Magna A Neque Industries	04 14 44 64 62	risus.donec.nibh@icloud.org	Australia	https://wha



» Next I will provide the code, and a screenshot of its successful execution:

```
CREATE OR REPLACE EXTERNAL TABLE
`sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw`
OPTIONS (
  format='CSV',
  uris=['gs://bootcamp-data-analytics-public/ERP/transactions.csv'],
  field_delimiter=';');

CREATE OR REPLACE EXTERNAL TABLE
`sprint3-analytics-marc-ponce.sprint3_bronze.companies_raw`
(
  company_id STRING,
  company_name STRING,
  phone STRING,
  email STRING,
  country STRING,
  website STRING
)
OPTIONS (
  format='CSV',
  uris=['gs://bootcamp-data-analytics-public/ERP/companies.csv'],
  skip_leading_rows=1
);

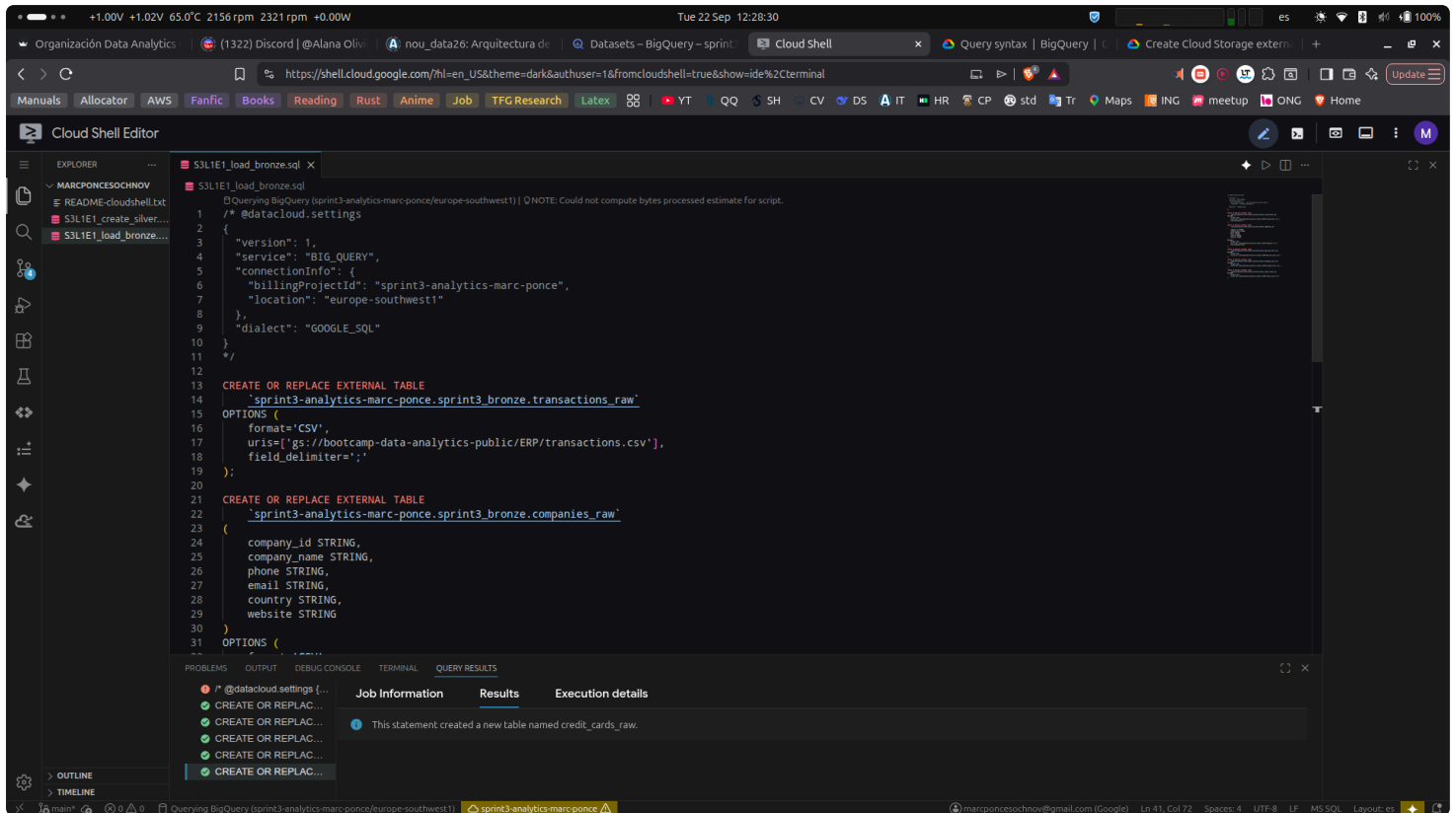
CREATE OR REPLACE EXTERNAL TABLE
`sprint3-analytics-marc-ponce.sprint3_bronze.american_users_raw`
OPTIONS (
  format='CSV',
  uris=['gs://bootcamp-data-analytics-public/CRM/american_users.csv']
);
```

CREATE OR REPLACE EXTERNAL TABLE

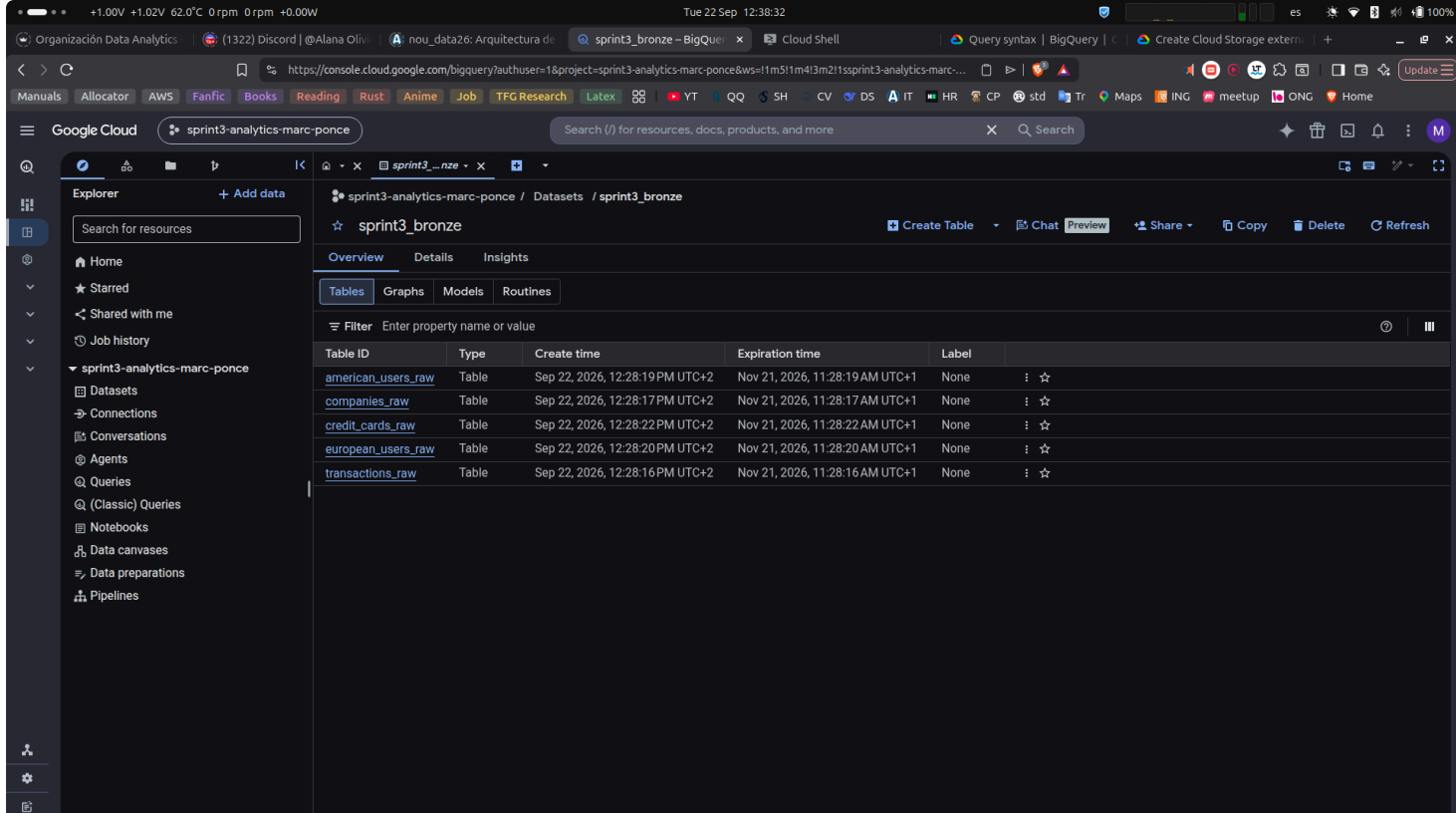
```
`sprint3-analytics-marc-ponce.sprint3_bronze.european_users_raw`  
OPTIONS (  
  format='CSV',  
  uris=['gs://bootcamp-data-analytics-public/CRM/european_users.csv']  
);
```

CREATE OR REPLACE EXTERNAL TABLE

```
`sprint3-analytics-marc-ponce.sprint3_bronze.credit_cards_raw`  
OPTIONS (  
  format='CSV',  
  uris=['gs://bootcamp-data-analytics-public/CRM/credit_cards.csv']  
);
```



» Lastly we check that the tables are in place:

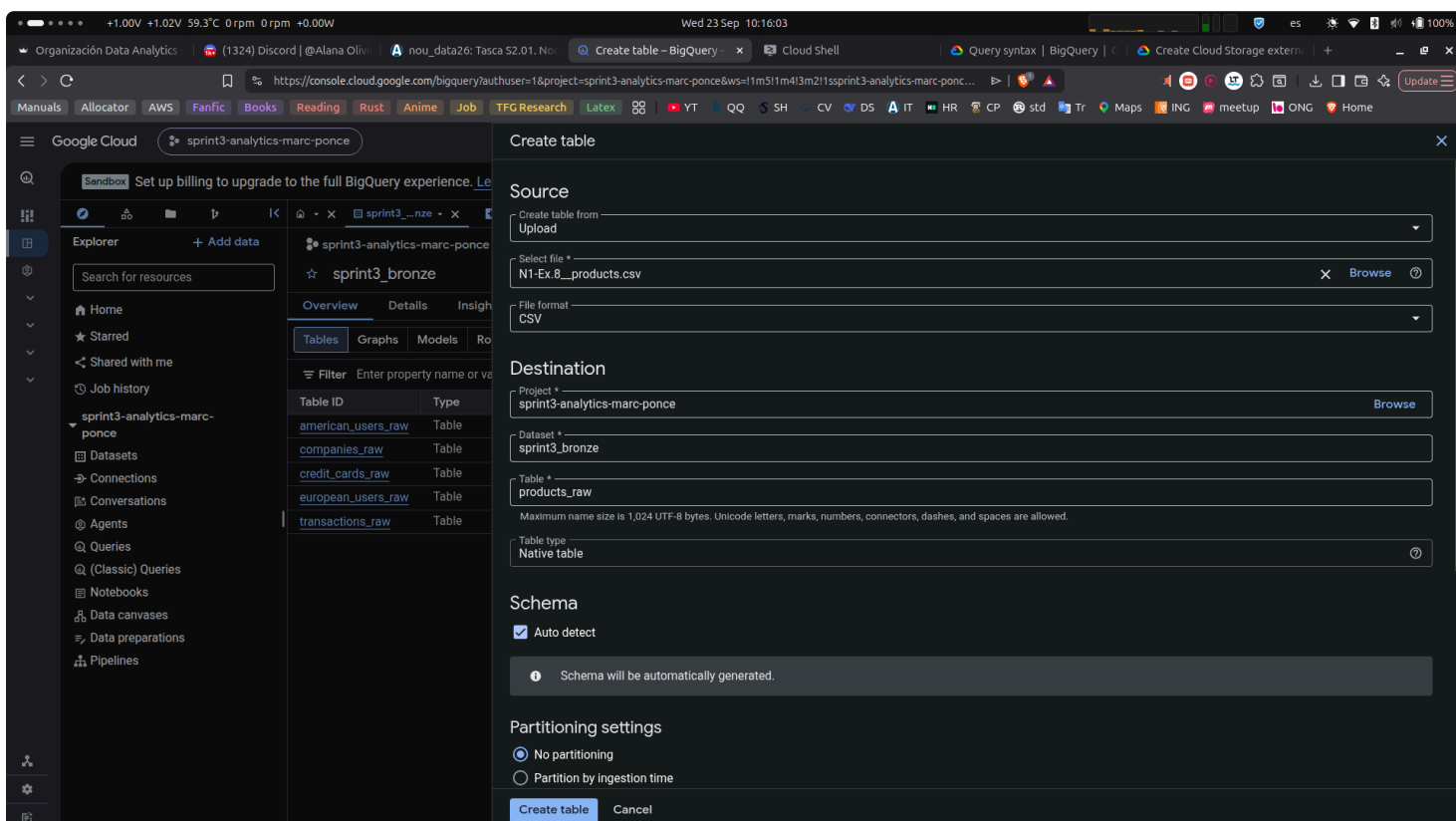


Exercici 3: Càrrega de Dades Locals (Upload)

Falta el catàleg de **Productes**. Aquesta informació no és al Data Lake, utilitza el fitxer *products.csv* de l'Sprint 2.

- Carrega aquest fitxer manualment ("Upload") al dataset `sprint3_bronze` i crea la taula `products_raw`.
- Observa que aquesta serà l'única **Taula Nativa** de moment.

» Upload the `.csv` to `sprint3_bronze.products_raw`



» Confirm its presence in `sprint3_bronze`.

Table ID	Type	Create time	Expiration time	Label	
american_users_raw	Table	Sep 22, 2026, 12:28:19 PM UTC+2	Nov 21, 2026, 11:28:19 AM UTC+1	None	☆
companies_raw	Table	Sep 22, 2026, 12:28:17 PM UTC+2	Nov 21, 2026, 11:28:17 AM UTC+1	None	☆
credit_cards_raw	Table	Sep 22, 2026, 12:28:22 PM UTC+2	Nov 21, 2026, 11:28:22 AM UTC+1	None	☆
european_users_raw	Table	Sep 22, 2026, 12:28:20 PM UTC+2	Nov 21, 2026, 11:28:20 AM UTC+1	None	☆
products_raw	Table	Sep 23, 2026, 10:19:42 AM UTC+2	Nov 22, 2026, 9:19:42 AM UTC+1	None	☆
transactions_raw	Table	Sep 22, 2026, 12:28:16 PM UTC+2	Nov 21, 2026, 11:28:16 AM UTC+1	None	☆

Exercici 4: Arquitectura i Rendiment. Materialització de Dades (Assistit per IA)

Escenari:

Les consultes sobre el Data Lake van lentes. Has de demostrar al teu mànager la diferència de rendiment entre treballar amb fitxers externs i treballar amb taules natives de BigQuery.

a) Materialització de Dades (Assistit per IA)

Tasca:

Crea una nova taula anomenada `sprint3_bronze.transactions_raw_native` que sigui una còpia exacta de la teva taula externa `transactions_raw`.

Requisit de "AI-Pair Programming":

No escriguis el codi SQL manualment. Utilitza l'Assistent d'IA (Gemini) integrat a BigQuery (o el xat lateral) per generar la sentència.

👉 **El Teu Rol:** Ets el responsable de validar que el codi que et dona la IA és correcte abans d'executar-lo.

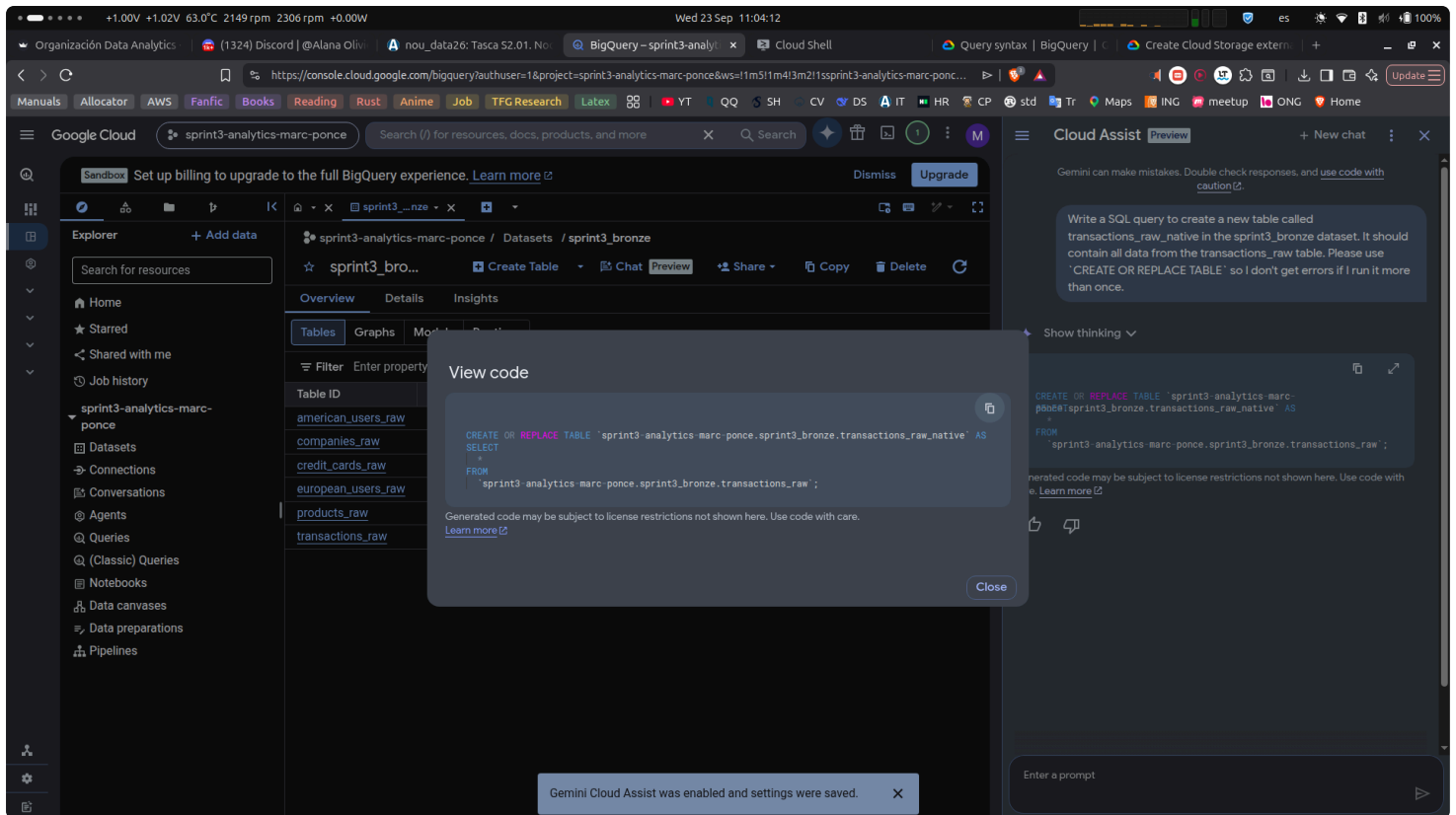
Prompt Suggestir:

"Write a SQL query to create a new table called `transactions_raw_native` in the `sprint3_bronze` dataset. It should contain all data from the `transactions_raw` table. Please use `CREATE OR REPLACE TABLE` so I don't get errors if I run it more than once."

👉 Entrega:

Una captura de pantalla on es vegi:

- El prompt que vas escriure a la IA.

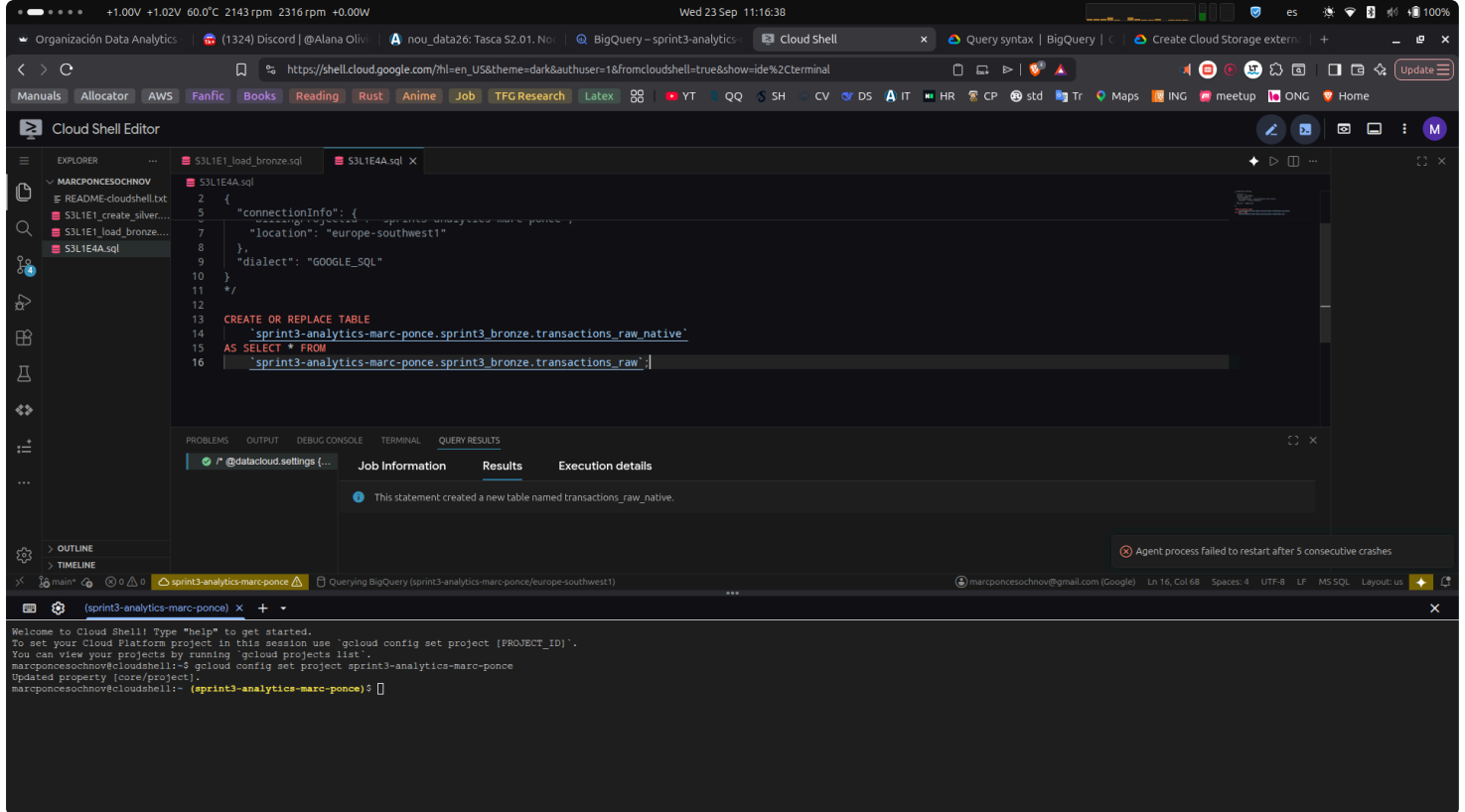


- El codi SQL que la IA va generar.

```
CREATE OR REPLACE TABLE
`sprint3-analytics-marc-ponce.sprint3-bronze.transactions_raw_native`
AS SELECT * FROM
`sprint3-analytics-marc-ponce.sprint3-bronze.transactions_raw`;
```

» So far I can see the code makes sense: we are just creating a table that is not `EXTERNAL`, which I am quite sure means it's `native`. Created from a standard `SELECT *`, the names are not mangled, so I deem this code correct.

- El resultat de l'execució ("This statement created a new table...").



» "This statement created a new table named transactions_raw_native."

b) Auditoria de Costos

Tasca:

Dissenyar una prova per comparar el cost de llegir una sola columna (ex: id) a la taula externa vs. la taula nativa. Utilitza els Bytes processats i els Bytes facturats per fer la comparació.

```
SELECT t.id
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw` AS t;

SELECT t.id
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw_native` AS t;
```

- Quan "pesa" la consulta a transactions_raw ?

Cloud Shell Editor interface showing a SQL query execution. The query is:

```
SELECT t.id
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw` AS t;

SELECT t.id
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw_native` AS t;
```

The execution details are as follows:

Job ID	sprint3-analytics-marc-ponce:europa-southwest1.script_job_77eb85f86e4a574b55d2f41c69a23475_0
User	MarcPonceSochnov@gmail.com
Location	europa-southwest1
Creation time	Sep 23, 2026, 11:31:16AM
Start time	Sep 23, 2026, 11:31:16AM
End time	Sep 23, 2026, 11:31:17AM
Duration	1.194 sec
Bytes processed	13220706
Bytes billed	13631488
Slot milliseconds	1398
Job priority	INTERACTIVE
Use legacy SQL	false
Destination table	sprint3-analytics-marc-ponce_5344bfc8b113c946a01148977895b2373c7bdc09.anon0c973f1bd4c38755f688fd574d8d4d6871e246aed3af1d643d98bbf7d7cac223
Query	SELECT t.id FROM `sprint3-analytics-marc-ponce.sprint3_...
Labels	

Processed	Billed
13.220 MB	13.631 MB

- Quan "pesa" la mateixa consulta a `transactions_raw_native` ?

Cloud Shell Editor interface showing a SQL query execution. The query is:

```
SELECT t.id
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw` AS t;

SELECT t.id
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw_native` AS t;
```

The execution details are as follows:

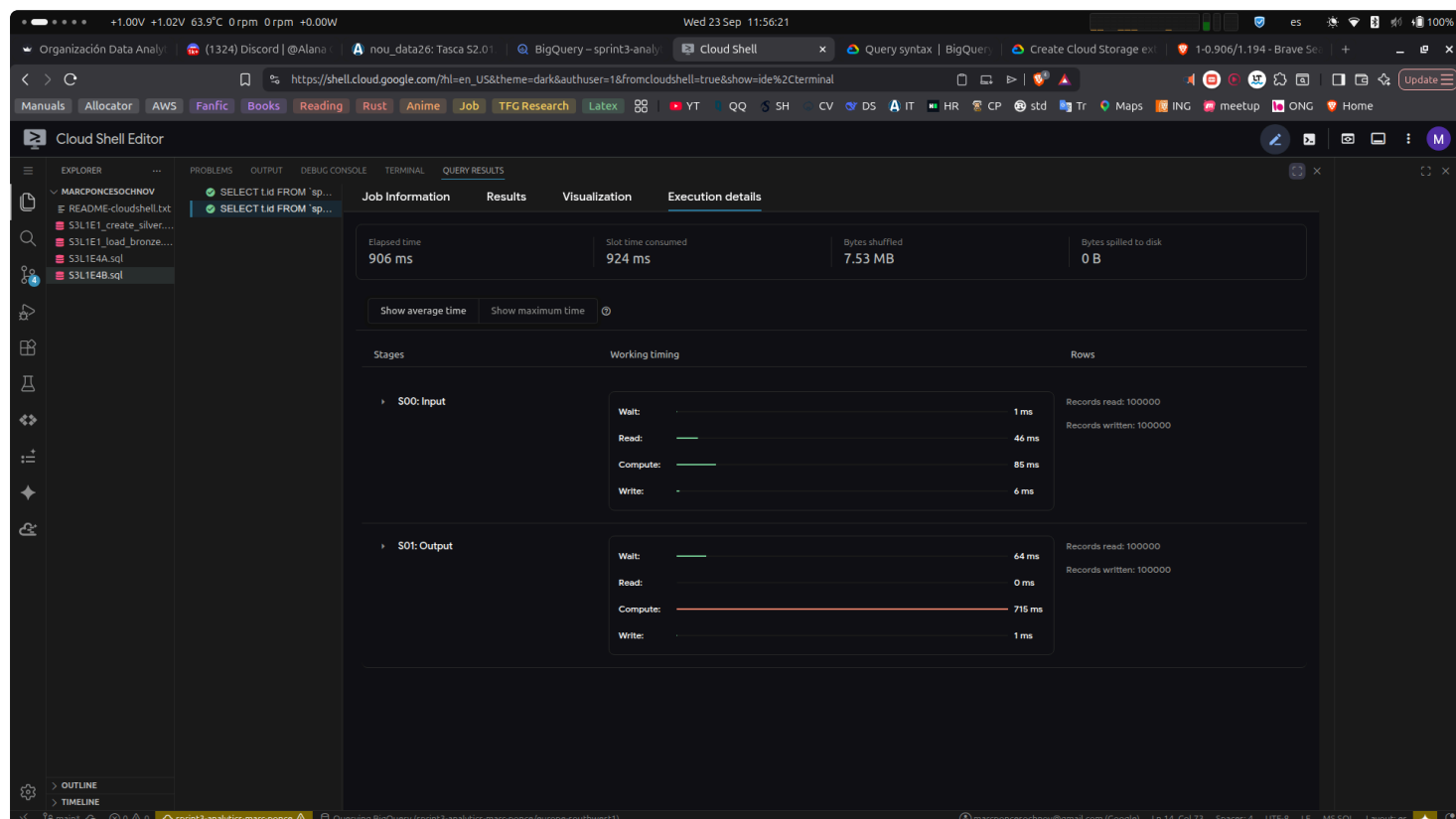
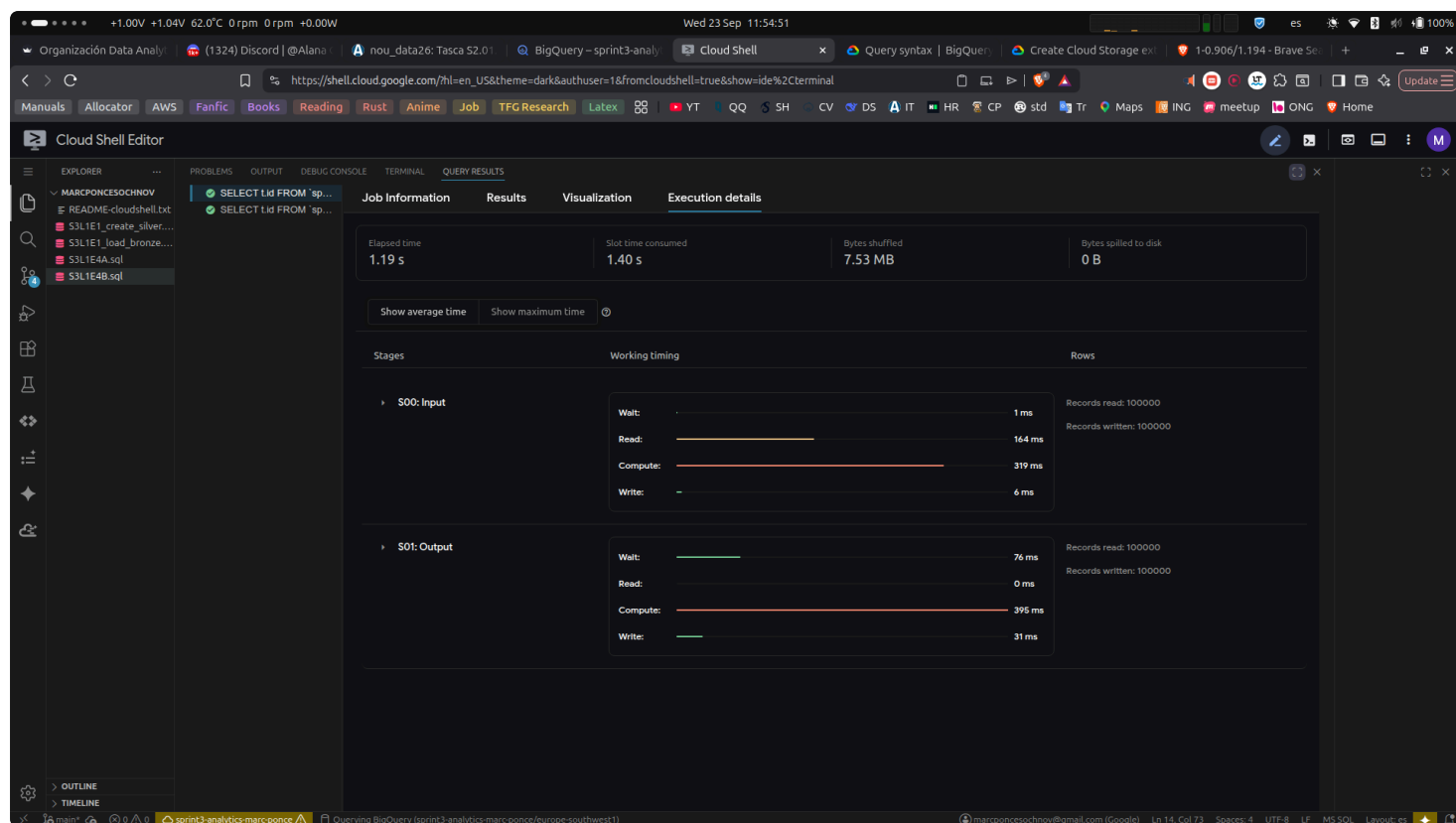
Job ID	sprint3-analytics-marc-ponce:europa-southwest1.script_job_ba4452cee0b4e8244baf1dd1678248c_1
User	MarcPonceSochnov@gmail.com
Location	europa-southwest1
Creation time	Sep 23, 2026, 11:31:18AM
Start time	Sep 23, 2026, 11:31:18AM
End time	Sep 23, 2026, 11:31:19AM
Duration	0.906 sec
Bytes processed	3800000
Bytes billed	10485760
Slot milliseconds	924
Job priority	INTERACTIVE
Use legacy SQL	false
Destination table	sprint3-analytics-marc-ponce_5344bfc8b113c946a01148977895b2373c7bdc09.anon8a6b6b1d496440ce2bf74236e5d783108092940d751cc6369134530412b85995
Query	SELECT t.id FROM `sprint3-analytics-marc-ponce.sprint3_...
Labels	

Processed	Billed
3.800 MB	10.485 MB

Entrega:

Captures del validador mostrant la diferència i una conclusió tècnica.

» The difference in billing is of 3MB , which makes the native table 23% cheaper. Comparing the execution times, it is also 24% faster.



» We can also observe that the compute and read in S00 varies much more than some 20-25% , which might indicate a great difference in energy consumption.

c) El perill del LIMIT

El LIMIT estalvia dades processades a BIGQUERY?



Fes un **LIMIT 10** sobre la taula `transactions_raw_native` y la taula `transactions_raw`

```
SELECT t.id
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw` AS t
LIMIT 10;
```

```
SELECT t.id
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw_native` AS t
LIMIT 10;
```

✅ **Comprovació:** Mira el "Dry Run". Ha baixat el cost respecte a no posar **LIMIT** ? Explica breument per què això és una trampa per a principiants.

The screenshot shows the Google Cloud Shell Editor interface. The left sidebar contains an Explorer view with a file tree for 'MARCPONCESOCHNOV'. The main editor area displays two SQL queries. The first query selects from 'transactions_raw' with a LIMIT of 10. The second query selects from 'transactions_raw_native' with a LIMIT of 10. Below the queries, the 'QUERY RESULTS' tab is active, showing the output of a dry run command: 'bq query --dry_run < S3L1E4C.sql'. The output indicates that the query was successfully validated and that it would process a lower bound of 3800000 bytes of data.

» The dry run yields a lower bound of 3.8MB. Lower bound means the minimum amount of bytes that it expects will be processed. There is no upper bound. I have to mention that it doesn't seem to account for the `EXTERNAL TABLE` query, and the `LIMIT` does not seem to cut off the bytes that it expects to process.

Cloud Shell Editor interface showing a query execution job. The query is `SELECT t.id FROM `sprint3-analytics-marc-ponce.sprint3_...``. The job information is as follows:

Job ID	sprint3-analytics-marc-ponce:europa-southwest1_script_job_01689bb76aa80f727b5a351b2c68655_0
User	MarcPonceSochnov@gmail.com
Location	europa-southwest1
Creation time	Sep 23, 2026, 12:17:51 PM
Start time	Sep 23, 2026, 12:17:52 PM
End time	Sep 23, 2026, 12:17:52 PM
Duration	0.537 sec
Bytes processed	1471
Bytes billed	10485760
Slot milliseconds	204
Job priority	INTERACTIVE
Use legacy SQL	false
Destination table	sprint3-analytics-marc-ponce_53a4bfc8b113c946a01148977895b2373c7bdc09_anona4351b0e32edac00875cec689d572c50783294ca98d0872f98025ad7ed34797a
Query	SELECT t.id FROM `sprint3-analytics-marc-ponce.sprint3_...`
Labels	

Cloud Shell Editor interface showing the execution details of the query. The execution details are as follows:

Elapsed time	Slot time consumed	Bytes shuffled	Bytes spilled to disk
537 ms	204 ms	410 B	0 B

Stages: S00: Output

Working timing	Rows
Wait: 1 ms	Records read: 11
Read: 119 ms	Records written: 10
Compute: 20 ms	
Write: 3 ms	

» With `transactions_raw` we can observe that the `LIMIT` did cut the amount of bytes processed. In fact, it can be further seen that only 11 records were processed. But we were billed 10MB regardless.

» With `transactions_raw_native` we are billed the same 10MB . It is faster, and most likely uses less energy. The amount of bytes processed doesn't differ from the `dry-run` lower bound estimation, and it reads and processes all 10000 records.

» `LIMIT` does make the numbers change, but it does not pause the machine after reading the first ten rows. Consulting an AI search engine I have come to the following information:

“Although the `LIMIT` clause limits the rows that a query produces, it doesn't limit the amount of data processed by that query.”

» Apparently Google's documentation explicitly recommends not to use `LIMIT` as a cost control mechanism for non-clustered tables. Secondly, BigQuery has a minimum billing amount of 10MB for

on-demand queries.

» For the `EXTERNAL TABLE`, BigQuery is querying the underlying external data rather than Google's own optimized native storage. Conceptually, this means that the `.csv` file must be read and parsed to produce the tabular data required by the query, rather than being accessed through BigQuery's native columnar storage.

Exercici 5: Adaptació de Sintaxi "Reporting"

El teu cap vol saber quins van ser els **5 dies amb més ingressos de l'any 2021**.

🦾 Repte:

Probablement, el camp `timestamp` és un `STRING`. Hauràs d'investigar funcions de BigQuery (`SUBSTR`, `CAST`, `PARSE_TIMESTAMP`) per filtrar l'any i agrupar per data correctament.

» The `timestamp` field is, in fact, not a `string`. BigQuery had loaded it into the proper `TIMESTAMP` form.

The screenshot shows a Google Cloud Shell interface. The terminal window displays a BigQuery query and its results. The query is as follows:

```
SELECT
  DATE(t.timestamp) AS t_day,
  SUM(t.amount) AS day_total
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw_native` AS t
GROUP BY t_day
ORDER BY day_total DESC
LIMIT 5;
```

The results are displayed in a table with two columns: `t_day` and `day_total`. The results are as follows:

Row	t_day	day_total
1	2022-12-13	14337.440000000002
2	2019-11-18	13591.320000000002
3	2023-02-20	13332.589999999997
4	2017-12-20	13318.429999999997
5	2019-03-18	12680.949999999993

```
SELECT
  DATE(t.timestamp) AS t_day,
  SUM(t.amount) AS day_total
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw_native` AS t
GROUP BY t_day
ORDER BY day_total DESC
LIMIT 5;
```

Exercici 6: Consultes Complexes

Necessitem un informe que creui dades.



Llista el nom, país i data de les transaccions realitzades per empreses que van fer operacions entre 100 i 200 euros en alguna d'aquestes dates: 29-04-2015, 20-07-2018 o 13-03-2024.

```
SELECT
    eu.name,
    c.country,
    DATE(t.timestamp) AS t_day
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw_native` AS t
JOIN `sprint3-analytics-marc-ponce.sprint3_bronze.companies_raw` AS c
ON t.business_id = c.company_id
JOIN `sprint3-analytics-marc-ponce.sprint3_bronze.european_users_raw` AS eu
ON t.user_id = eu.id
WHERE t.amount BETWEEN 100 AND 200
AND t_day in ('29-04-2015', '20-07-2018', '13-03-2024')
UNION ALL
SELECT
    au.name,
    c.country,
    DATE(t.timestamp) AS t_day
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw_native` AS t
JOIN `sprint3-analytics-marc-ponce.sprint3_bronze.companies_raw` AS c
ON t.business_id = c.company_id
JOIN `sprint3-analytics-marc-ponce.sprint3_bronze.american_users_raw` AS au
ON t.user_id = au.id
WHERE t.amount BETWEEN 100 AND 200
AND t_day in ('29-04-2015', '20-07-2018', '13-03-2024');
```

```
Mon 28 Sep 10:04:25
share.streamlit.io | Discord | #reskilling_data | nou_data26: Tasca S2.01 | BigQuery - sprint3-analy | Cloud Shell
https://shell.cloud.google.com/terminal?hl=en_US&theme=dark&authuser=1&fromcloudshell=true&show=ide%2Cterminal
Manuals | Allocator | AWS | Fanfic | Books | Reading | Rust | Anime | Job | TFG Research | Latex | YT | QQ | SH | CV | DS | IT | HR | CP | std | Tr | Maps | ING | meetup | ONG | Home
Cloud Shell Editor
EXPLORER
MARCPONCESOCHNOV
  README-cloudshell.txt
  S3L1E1_create_silver...
  S3L1E1_load_bronze...
  S3L1E4A.sql
  S3L1E4B.sql
  S3L1E4C.sql
  S3L1E5.sql
  S3L1E6.sql
S3L1E6.sql
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SELECT
    eu.name,
    c.country,
    DATE(t.timestamp) AS t_day
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw_native` AS t
JOIN `sprint3-analytics-marc-ponce.sprint3_bronze.companies_raw` AS c
ON t.business_id = c.company_id
JOIN `sprint3-analytics-marc-ponce.sprint3_bronze.european_users_raw` AS eu
ON t.user_id = eu.id
WHERE t.amount BETWEEN 100 AND 200
AND DATE(t.timestamp) in ('2015-04-29', '2018-07-20', '2024-03-13')
UNION ALL
SELECT
    au.name,
    c.country,
    DATE(t.timestamp) AS t_day
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw_native` AS t
JOIN `sprint3-analytics-marc-ponce.sprint3_bronze.companies_raw` AS c
ON t.business_id = c.company_id
JOIN `sprint3-analytics-marc-ponce.sprint3_bronze.american_users_raw` AS au
ON t.user_id = au.id
WHERE t.amount BETWEEN 100 AND 200
AND DATE(t.timestamp) in ('2015-04-29', '2018-07-20', '2024-03-13')
;
Querying BigQuery (sprint3-analytics-marc-ponce/europe-southwest1)
sprint3-analytics-marc-ponce
marcpocesochnov@cloudshell:~$ bq query --location=europe-southwest1 --dry-run < S3L1E6.sql
Query successfully validated. Assuming the tables are not modified, running this query will process lower bound of 3200000 bytes of data.
marcpocesochnov@cloudshell:~$
```

Mon 28 Sep 10:06:08

share.streamlit.io | Discord | #reskilling_data | nou_data26: Tasca S2.01 | BigQuery - sprint3-analy... | cloud Shell

https://shell.cloud.google.com/?hl=en_US&theme=dark&authuser=1&fromcloudshell=true&show=ide%2Cterminal

Manually | Allocator | AWS | Fanfic | Books | Reading | Rust | Anime | Job | TFG Research | Latex | YT | QQ | SH | CV | DS | IT | HR | CP | std | Tr | Maps | ING | meetup | ONG | Home

Cloud Shell Editor

EXPLORER

- MARCPONCESOCHNOV
 - README-cloudshell.txt
 - S3L1E1_create_silver...
 - S3L1E1_load_bronze...
 - S3L1E4A.sql
 - S3L1E4B.sql
 - S3L1E4C.sql
 - S3L1E5.sql
 - S3L1E6.sql

```
13 SELECT
14   eu.name,
15   c.country,
16   DATE(t.timestamp) AS t_day
17 FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw_native` AS t
18 JOIN `sprint3-analytics-marc-ponce.sprint3_bronze.companies_raw` AS c
19 ON t.business_id = c.company_id
20 JOIN `sprint3-analytics-marc-ponce.sprint3_bronze.european_users_raw` AS eu
21 ON t.user_id = eu.id
22 WHERE t.amount BETWEEN 100 AND 200
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL QUERY RESULTS

@datacloud.settings [...] Job Information Results Visualization Execution details

Row	name	country	t_day
1	Xijhaa	United States	2018-07-20
2	Walter	United States	2015-04-29
3	Fvoxuw	New Zealand	2018-07-20
4	Alka	Canada	2015-04-29
5	Gowerg	Belgium	2018-07-20
6	Dbvfmj	United Kingdom	2015-04-29
7	Morrek	France	2015-04-29
8	Rbrhvu	United Kingdom	2024-03-13
9	Ygcgja	United Kingdom	2015-04-29
10	Emmwaz	Italy	2018-07-20

Items per page: 50 1 - 19 of 19

main* 0 0 0 Querying BigQuery [sprint3-analytics-marc-ponce/europe-southwest1] sprint3-analytics-marc-ponce

Mon 28 Sep 10:07:08

share.streamlit.io | Discord | #reskilling_data | nou_data26: Tasca S2.01 | BigQuery - sprint3-analy... | cloud Shell

https://shell.cloud.google.com/?hl=en_US&theme=dark&authuser=1&fromcloudshell=true&show=ide%2Cterminal

Manually | Allocator | AWS | Fanfic | Books | Reading | Rust | Anime | Job | TFG Research | Latex | YT | QQ | SH | CV | DS | IT | HR | CP | std | Tr | Maps | ING | meetup | ONG | Home

Cloud Shell Editor

EXPLORER

- MARCPONCESOCHNOV
 - README-cloudshell.txt
 - S3L1E1_create_silver...
 - S3L1E1_load_bronze...
 - S3L1E4A.sql
 - S3L1E4B.sql
 - S3L1E4C.sql
 - S3L1E5.sql
 - S3L1E6.sql

```
13 SELECT
14   eu.name,
15   c.country,
16   DATE(t.timestamp) AS t_day
17 FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw_native` AS t
18 JOIN `sprint3-analytics-marc-ponce.sprint3_bronze.companies_raw` AS c
19 ON t.business_id = c.company_id
20 JOIN `sprint3-analytics-marc-ponce.sprint3_bronze.european_users_raw` AS eu
21 ON t.user_id = eu.id
22 WHERE t.amount BETWEEN 100 AND 200
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL QUERY RESULTS

@datacloud.settings [...] Job Information Results Visualization Execution details

Job ID	sprint3-analytics-marc-ponce:europe-southwest1 job_Gwpc7xnVV1EXAfeq4QFFRc-NH7g
User	MarcPonceSochnov@gmail.com
Location	europe-southwest1
Creation time	Sep 28, 2026, 10:05:49 AM
Start time	Sep 28, 2026, 10:05:49 AM
End time	Sep 28, 2026, 10:05:50 AM
Duration	0.744 sec
Bytes processed	3802351
Bytes billed	41943040
Slot milliseconds	3434
Job priority	INTERACTIVE
Use legacy SQL	false
Destination table	sprint3-analytics-marc-ponce_53a4bfc8b113c946a01148977895b2373c7bdc09anon955362cb2d5755f8bc9b4506df5da02d115f9f7a917fb5ba2c463c15f82ee2f3
Query	/* @datacloud.settings { "version": 1, "service": "BIG_...
Labels	

main* 0 0 0 Querying BigQuery [sprint3-analytics-marc-ponce/europe-southwest1] sprint3-analytics-marc-ponce

Nivell 2: Neteja i Transformació (ELT)

Escenari:

Les dades brutes tenen problemes de qualitat (símbols de moneda, usuaris duplicats, dates incorrectes). Actua com a Data Engineer per netejar-ho tot.

Exercici 1: Neteja de Productes (Data Quality)

Crearem la capa neta ("Silver") per als productes. Tens la taula `sprint3_bronze.products_raw` carregada des del CSV. L'equip de Data Governance et

demana crear una taula de productes neta a `sprint3_silver.products_clean` que compleixi aquestes regles de qualitat:

- **Estandardització de Noms:** La columna original `id` és ambigua; reanomena-la a `product_id`. La columna `product_name` simplifica-la a `name`.
- **Neteja d'IDs:** El camp `warehouse_id` té el format antic "WH-4". Elimina el prefix "WH-" i converteix el valor a nombre enter (INT64).
- **Garantia de Preu:** Assegura't que `price` és un número `FLOAT64`, sense símbols de moneda.
- **Altres columnes:** Conserva el camp `weight` (pes) tal qual.

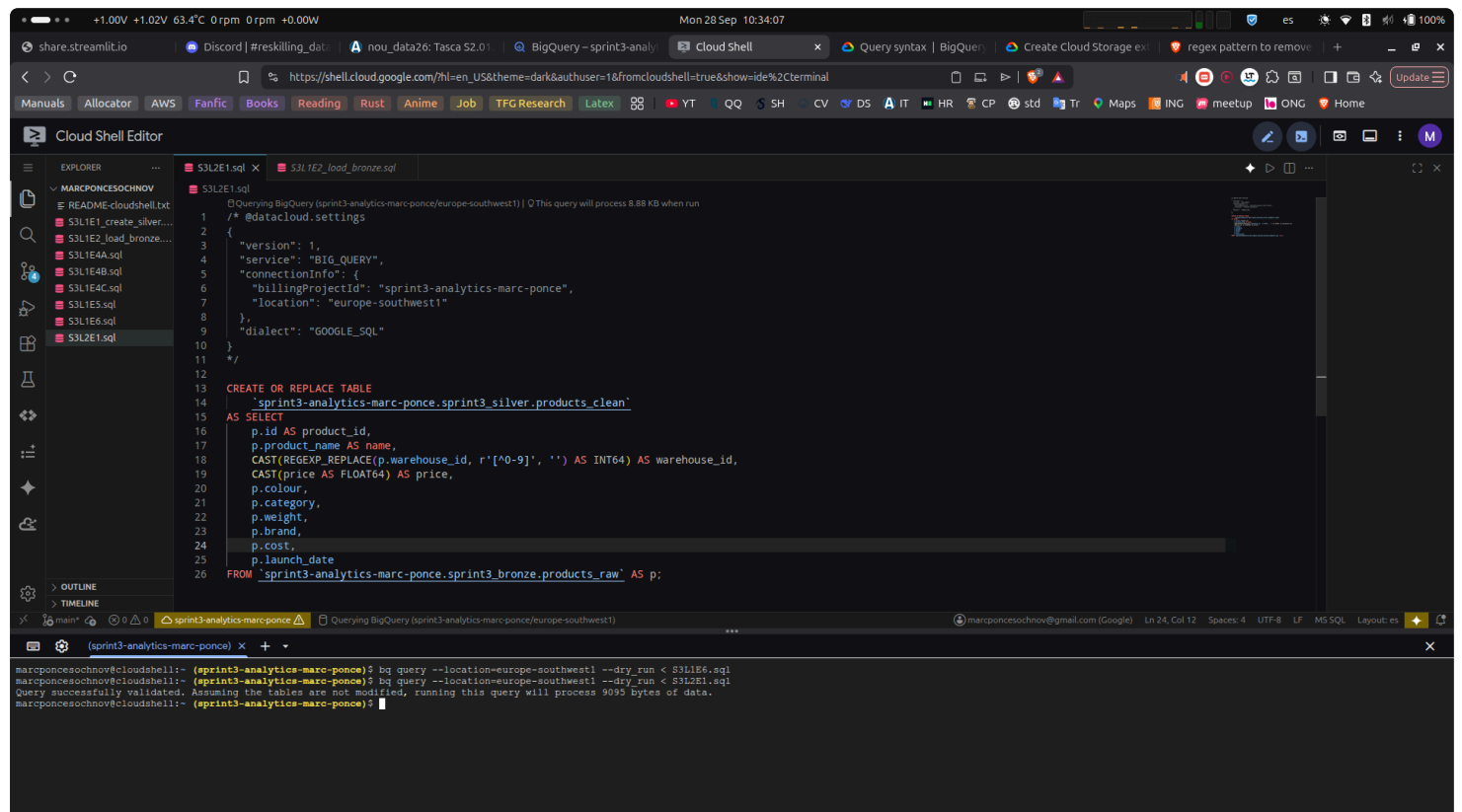
CREATE OR REPLACE TABLE

```
`sprint3-analytics-marc-ponce.sprint3_silver.products_clean`
```

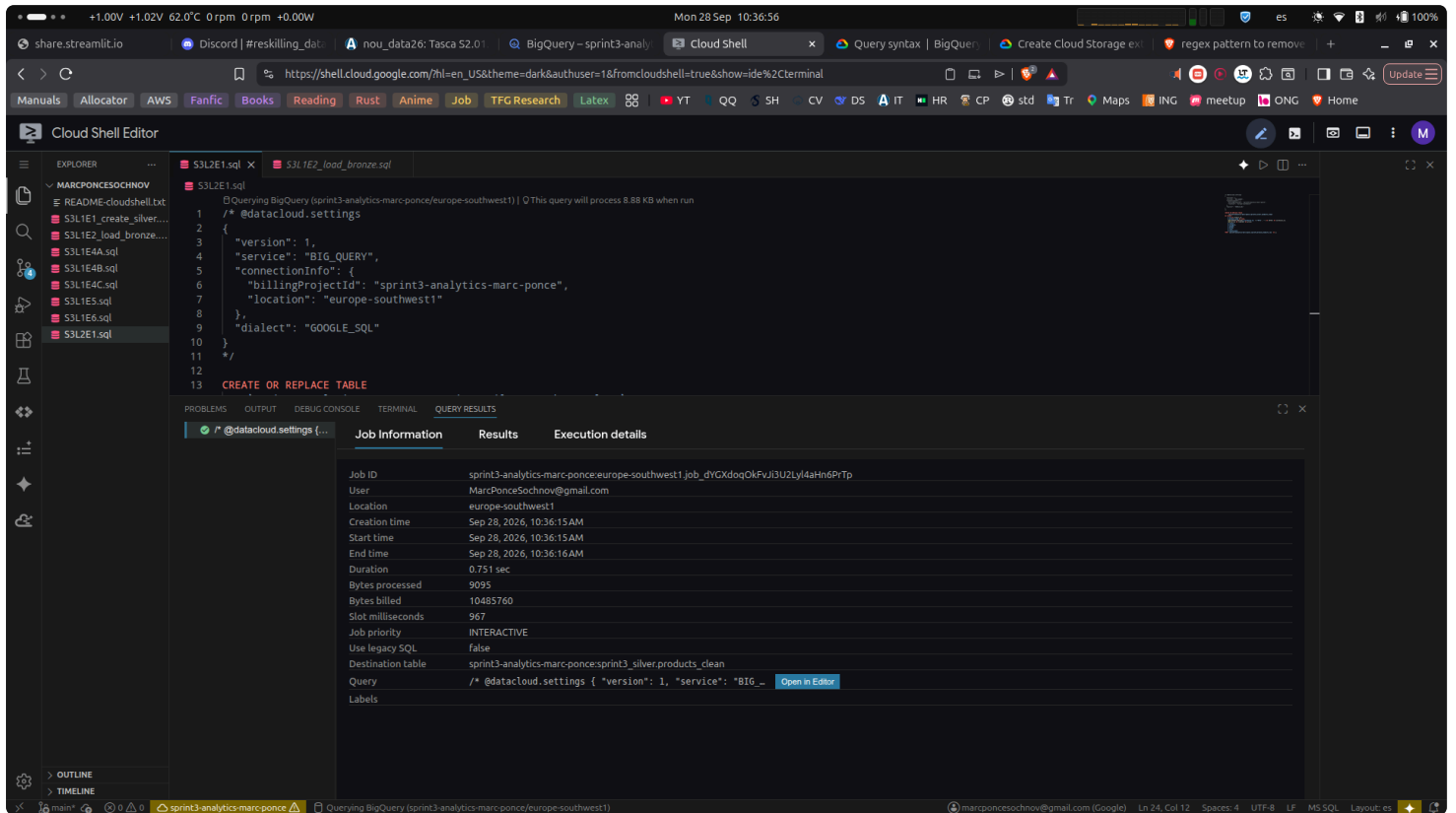
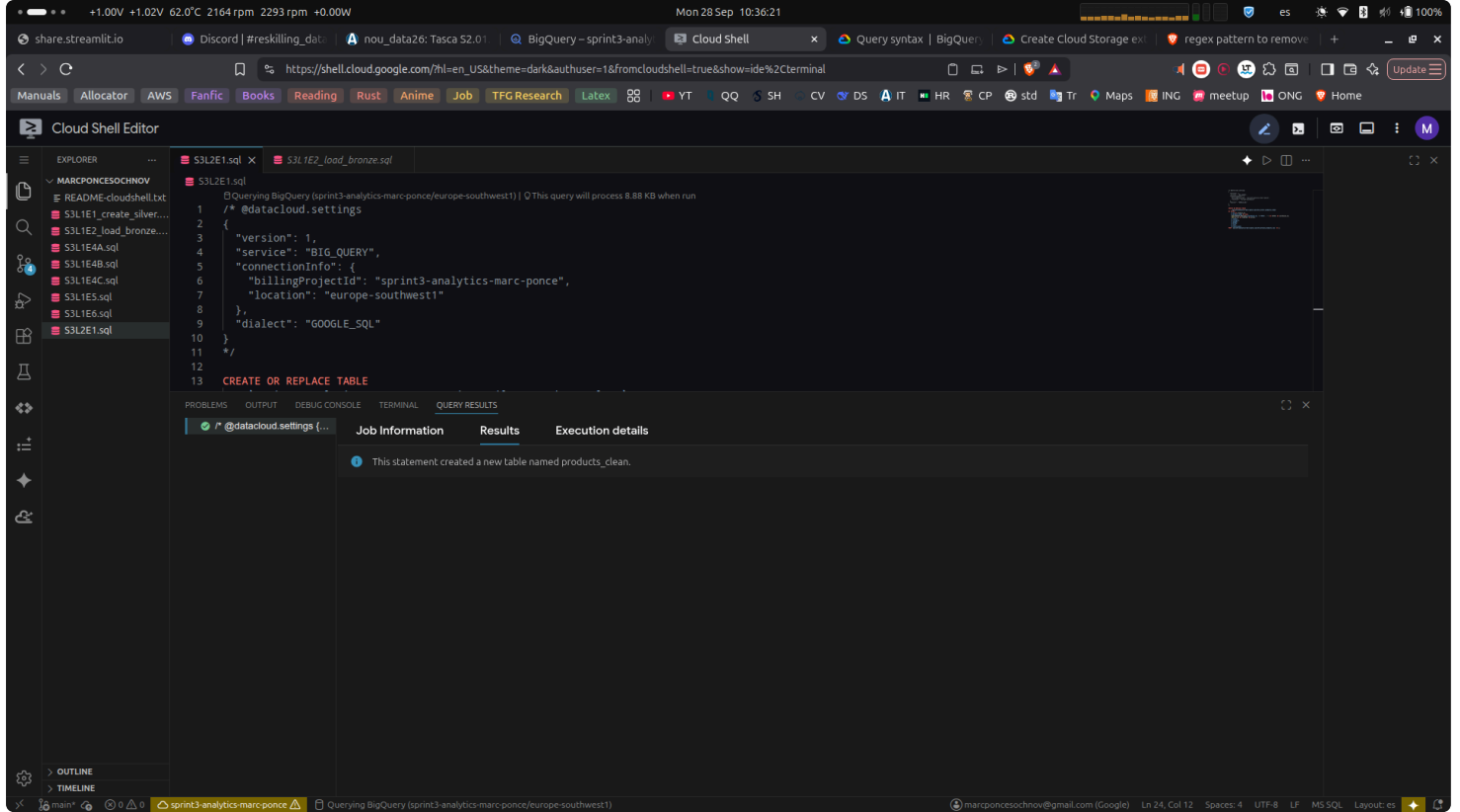
AS SELECT

```
p.id AS product_id,  
p.product_name AS name,  
CAST(REGEXP_REPLACE(p.warehouse_id, r'^[^\d-]*', '' ) AS INT64) AS warehouse_id,  
CAST(price AS FLOAT64) AS price,  
p.colour,  
p.category,  
p.weight,  
p.brand,  
p.cost,  
p.launch_date
```

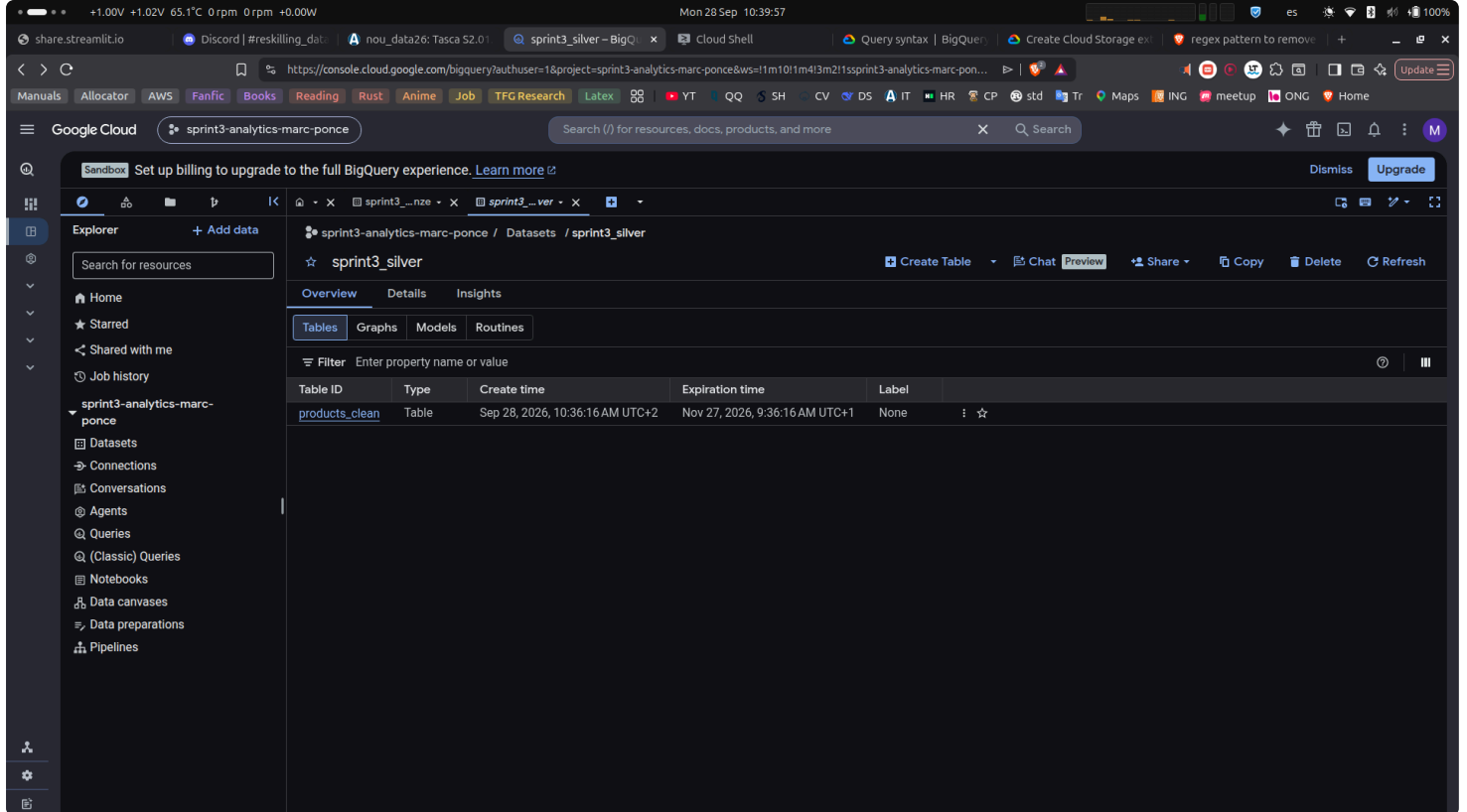
FROM `sprint3-analytics-marc-ponce.sprint3\_bronze.products\_raw` AS p;



```
Mon 28 Sep 10:34:07  
share.streamlit.io | Discord | #reskilling_data | nou_data26: Tasca S2.0 | BigQuery - sprint3-analy... | Cloud Shell | Query syntax | BigQuery | Create Cloud Storage ex... | regex pattern to remove | +  
https://shell.cloud.google.com/?hl=en_US&theme=dark&authuser=1&fromcloudshell=true&show-side%2Cterminal  
Manuals | Allocator | AWS | Fanfic | Books | Reading | Rust | Anime | Job | TFG Research | Latex | YT | QQ | SH | CV | DS | IT | HR | CP | std | Tr | Maps | ING | meetup | ONG | Home  
Cloud Shell Editor  
EXPLORER | S3L2E1.sql | S3L1E2_load_bronze.sql  
MARCPONCESOCHNOV  
README-cloudshell.txt  
S3L1E1_create_silver...  
S3L1E2_load_bronze...  
S3L1E4A.sql  
S3L1E4B.sql  
S3L1E4C.sql  
S3L1E5.sql  
S3L1E6.sql  
S3L2E1.sql  
S3L2E1.sql  
1 /* @datacloud.settings  
2 {  
3   "version": 1,  
4   "service": "BIG_QUERY",  
5   "connectionInfo": {  
6     "billingProjectId": "sprint3-analytics-marc-ponce",  
7     "location": "europe-southwest1"  
8   },  
9   "dialect": "GOOGLE_SQL"  
10 }  
11 */  
12  
13 CREATE OR REPLACE TABLE  
14 `sprint3-analytics-marc-ponce.sprint3_silver.products_clean`  
15 AS SELECT  
16   p.id AS product_id,  
17   p.product_name AS name,  
18   CAST(REGEXP_REPLACE(p.warehouse_id, r'^[^\d-]*', '' ) AS INT64) AS warehouse_id,  
19   CAST(price AS FLOAT64) AS price,  
20   p.colour,  
21   p.category,  
22   p.weight,  
23   p.brand,  
24   p.cost,  
25   p.launch_date  
26 FROM `sprint3-analytics-marc-ponce.sprint3_bronze.products_raw` AS p;  
Querying BigQuery (sprint3-analytics-marc-ponce/europe-southwest1) | This query will process 8.88 KB when run  
marcpocesochnov@cloudshell:~$ bq query --location=europe-southwest1 --dry_run < S3L1E6.sql  
marcpocesochnov@cloudshell:~$ bq query --location=europe-southwest1 --dry_run < S3L2E1.sql  
Query successfully validated. Assuming the tables are not modified, running this query will process 3090 bytes of data.  
marcpocesochnov@cloudshell:~$
```



» Last, we check that the table is where it should be:



Exercici 2: Creació de Transaccions Netes (Capa Silver)

Escenari:

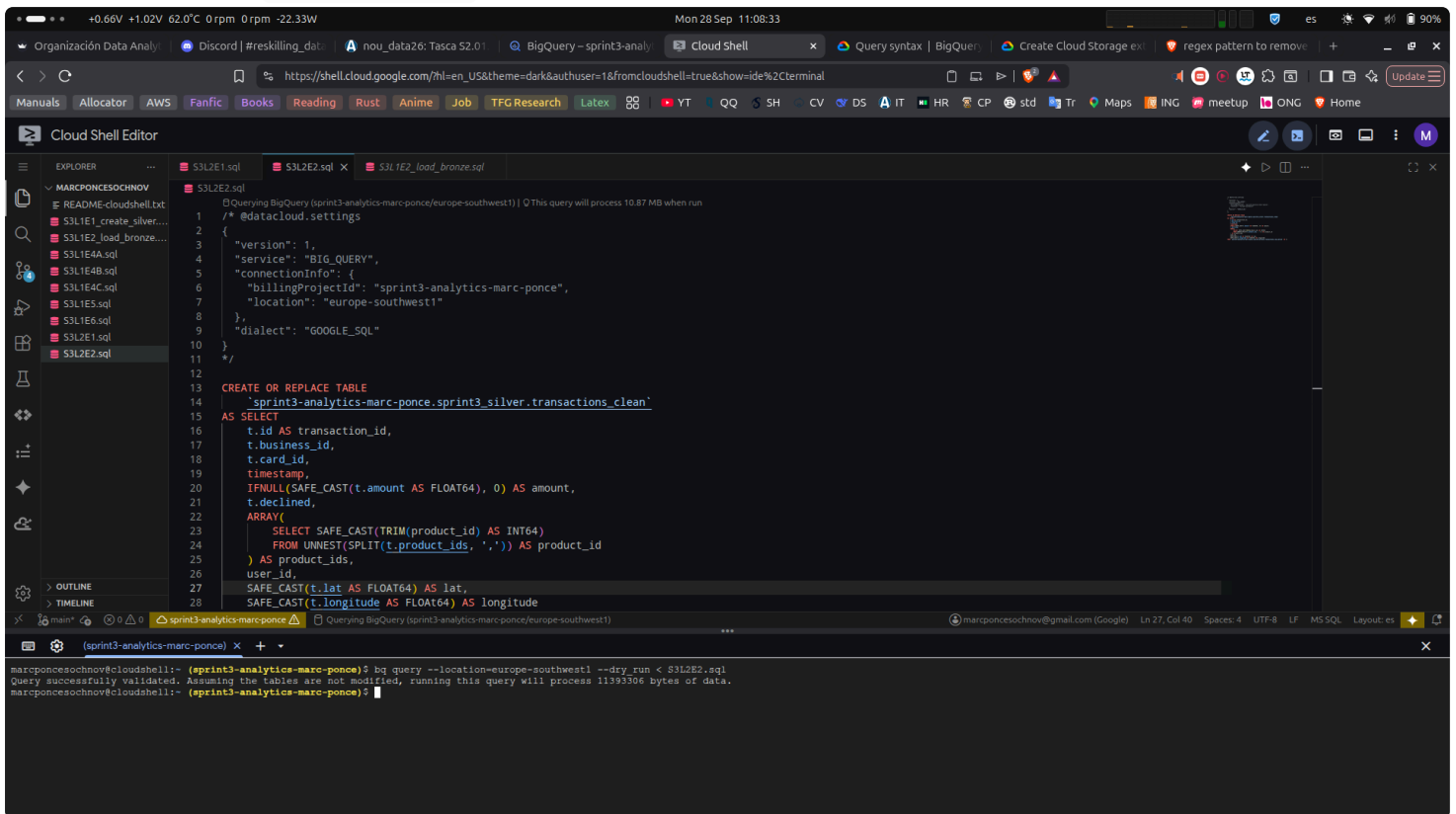
La taula `transactions_raw` (Bronze) és insegura: l'import (`amount`) podria tenir lletres, les coordenades podrien fallar i la data és un text difícil de filtrar. Crearem la taula definitiva `sprint3_silver.transactions_clean` que garanteixi el següent.

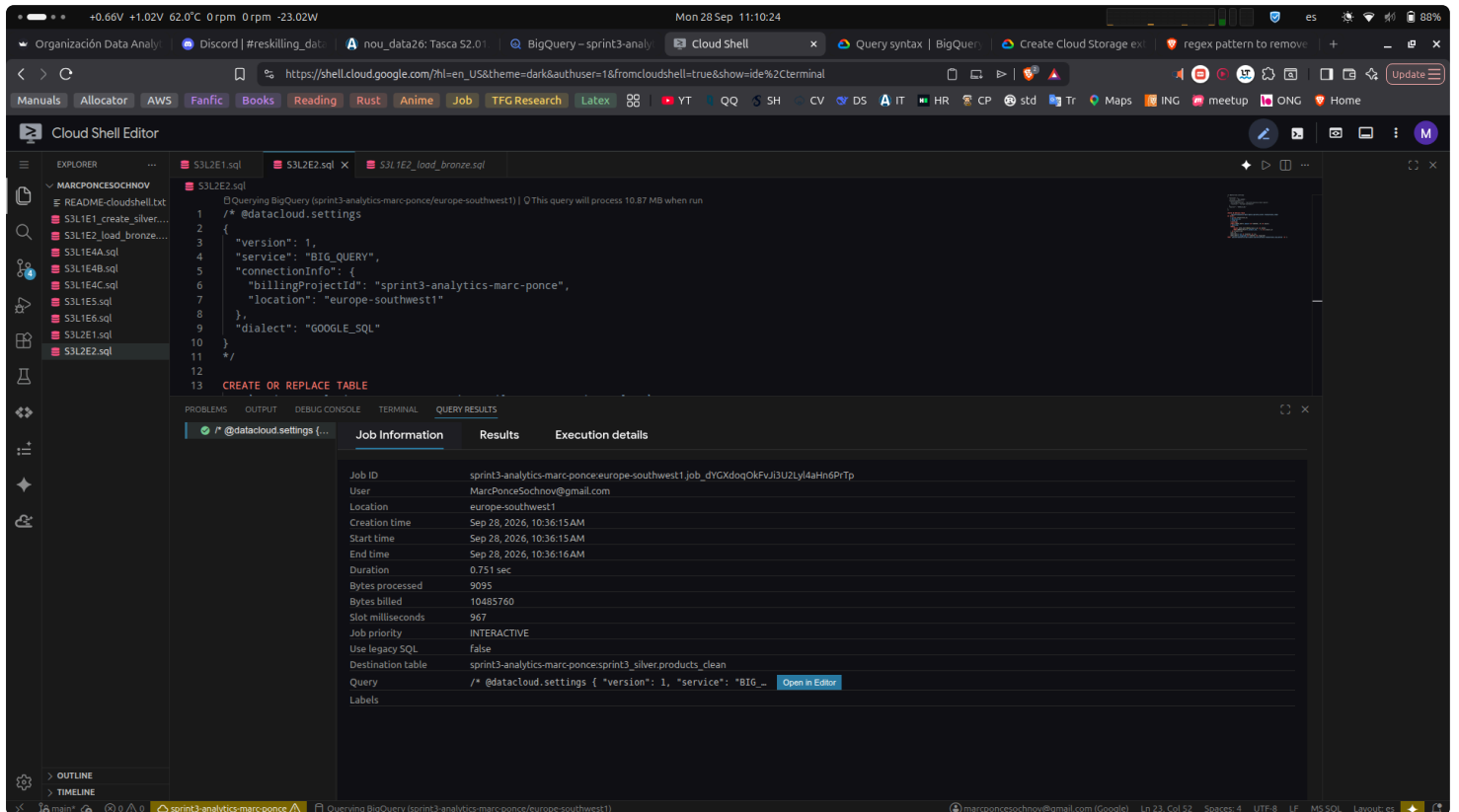
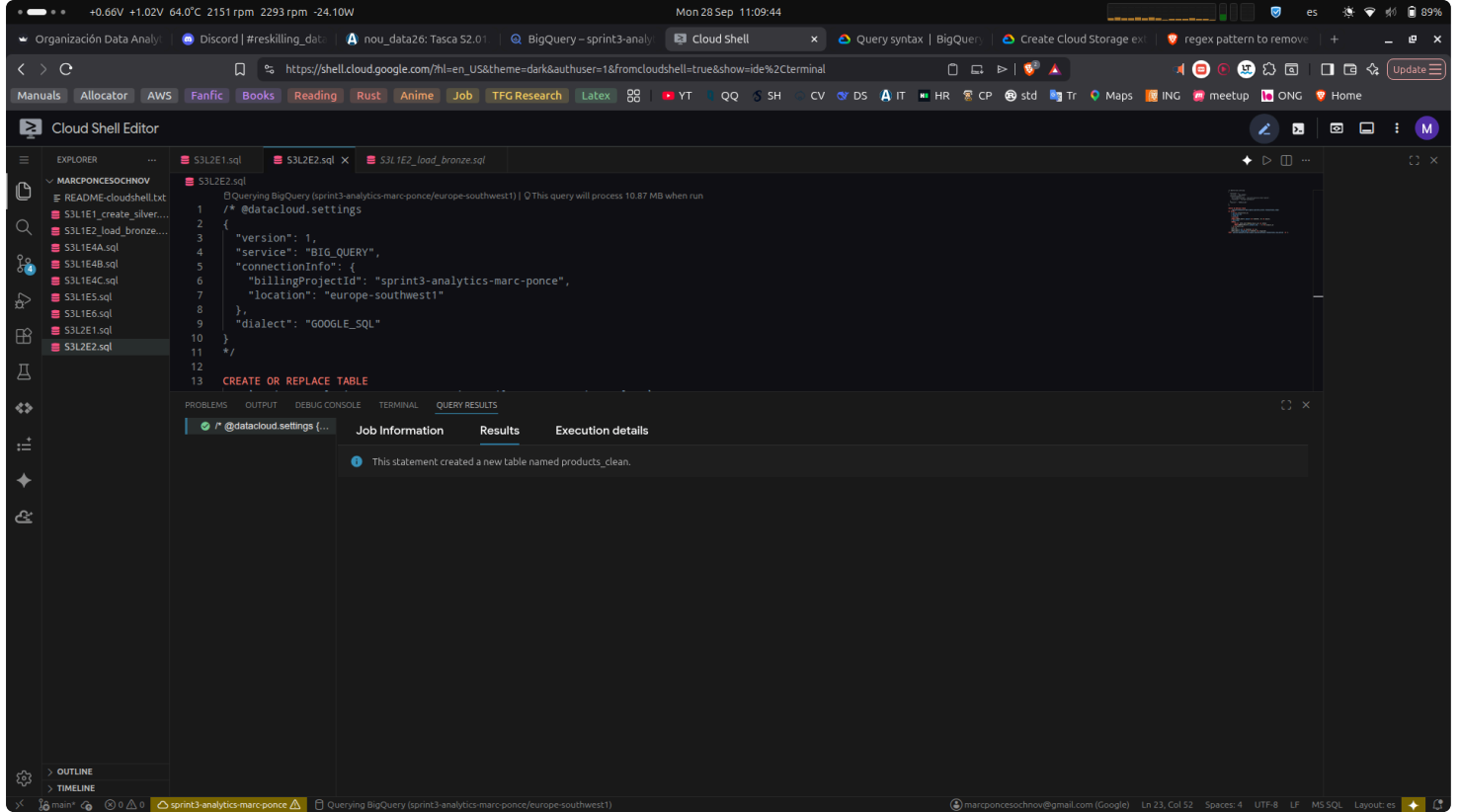
- **Estandardització de Noms:** La columna original `id` és ambigua; reanomena-la a `transaction_id`.
- **Robustesa en Imports:** Utilitza `SAFE_CAST` al camp `amount`. Si falla, substitueix-lo per 0 (usant `IFNULL`).
- **Dates Reals:** Converteix el camp `timestamp` (que és `STRING`) a tipus `TIMESTAMP` real.
- **Coordenades:** Assegura que `lat` i `longitude` siguin `FLOAT64` (utilitza `SAFE_CAST` per seguretat).
- **Desglossa els Productes:** Transforma la cadena de text `product_ids` en un `ARRAY` d'enters.
- **Input:** `"1, 2, 3"` `STRING`
- **Output:** `[1, 2, 3]` `ARRAY<INT>`
- **Resta de camps:** Mantén-los igual.

CREATE OR REPLACE TABLE

```
`sprint3-analytics-marc-ponce.sprint3_silver.transactions_clean`  
AS SELECT  
  t.id AS transaction_id,  
  t.business_id,  
  t.card_id,  
  timestamp,  
  IFNULL(SAFE_CAST(t.amount AS FLOAT64), 0) AS amount,  
  t.declined,  
  ARRAY(
```

```
SELECT SAFE_CAST(TRIM(product_id) AS INT64)
FROM UNNEST(SPLIT(t.product_ids, ',')) AS product_id
) AS product_ids,
user_id,
SAFE_CAST(t.lat AS FLOAT64) AS lat,
SAFE_CAST(t.longitude AS FLOAT64) AS longitude
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.transactions_raw_native` AS t;
```





» Finally we check that the table is in place:

The screenshot shows the Google Cloud BigQuery console interface. On the left is the Explorer sidebar with navigation options like Home, Starred, Shared with me, Job history, and various data sources. The main panel displays the schema for the 'transactions_clean' table within the 'sprint3_analytics_marc_ponce' dataset. The schema table lists columns with their respective types and modes. Notably, the 'product_ids' column is of type 'INTEGER' and mode 'REPEATED'. Other columns include 'transaction_id' (STRING, NULLABLE), 'business_id' (STRING, NULLABLE), 'card_id' (STRING, NULLABLE), 'timestamp' (TIMESTAMP, NULLABLE), 'amount' (FLOAT, NULLABLE), 'declined' (INTEGER, NULLABLE), 'user_id' (INTEGER, NULLABLE), 'lat' (FLOAT, NULLABLE), and 'longitude' (FLOAT, NULLABLE). Below the schema table are buttons for 'Edit schema', 'View row access policies', and 'Describe data'.

Field name	Type	Mode	Description	Key	Collation	Default Value	Data Governance Tags	Policy Tags	Data Policies
transaction_id	STRING	NULLABLE	-	-	-	-	-	-	-
business_id	STRING	NULLABLE	-	-	-	-	-	-	-
card_id	STRING	NULLABLE	-	-	-	-	-	-	-
timestamp	TIMESTAMP	NULLABLE	-	-	-	-	-	-	-
amount	FLOAT	NULLABLE	-	-	-	-	-	-	-
declined	INTEGER	NULLABLE	-	-	-	-	-	-	-
product_ids	INTEGER	REPEATED	-	-	-	-	-	-	-
user_id	INTEGER	NULLABLE	-	-	-	-	-	-	-
lat	FLOAT	NULLABLE	-	-	-	-	-	-	-
longitude	FLOAT	NULLABLE	-	-	-	-	-	-	-

» REPEATED means that the field holds zero or more values of the specified type (i.e., INT64 in this example).

Exercici 3: Unificació d'Usuaris (UNION)

Tenim els usuaris fragmentats per regió.

Tasca:

Crea la taula `sprint3_silver.users_combined`. Utilitza `UNION ALL` per unificar els usuaris dels EUA i Europa en una única llista mestra. Afegeix una columna calculada `origin` per saber d'on venen.

- **Estandardització de Noms:** La columna original `id` és ambigua; reanomena-la a `user_id`.

```
CREATE OR REPLACE TABLE
`sprint3-analytics-marc-ponce.sprint3_silver.users_combined`
AS SELECT
  eu.id AS user_id,
  eu.name,
  eu.surname,
  eu.phone,
  eu.email,
  eu.birth_date,
  eu.country,
  eu.city,
  eu.postal_code,
  eu.address,
  'EU' AS origin
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.european_users_raw` AS eu
UNION ALL
SELECT
  au.id AS user_id,
```

```

au.name,
au.surname,
au.phone,
au.email,
au.birth_date,
au.country,
au.city,
au.postal_code,
au.address,
'US' AS origin
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.american_users_raw` AS au;

```

Cloud Shell Editor

```

10
11
12
13
14 CREATE OR REPLACE TABLE
15   `sprint3-analytics-marc-ponce.sprint3_silver.users_combined`
16 AS SELECT
17   eu.id AS user_id,
18   eu.name,
19   eu.surname,
20   eu.phone,
21   eu.email,
22   eu.birth_date,
23   eu.country,
24   eu.city,
25   eu.postal_code,
26   eu.address,
27   'EU' AS origin
28 FROM `sprint3-analytics-marc-ponce.sprint3_bronze.european_users_raw` AS eu
29 UNION ALL
30 SELECT
31   au.id AS user_id,
32   au.name,
33   au.surname,
34   au.phone,
35   au.email,
36   au.birth_date,
37   au.country,
38   au.city,
39   au.postal_code,
40   au.address,
41   'US' AS origin
42 FROM `sprint3-analytics-marc-ponce.sprint3_bronze.american_users_raw` AS au;

```

marcponcesochnov@cloudshell:~\$ (sprint3-analytics-marc-ponce) bq query --dry-run < S3L2E3.sql
Query successfully validated. Assuming the tables are not modified, running this query will process lower bound of 0 bytes of data.
marcponcesochnov@cloudshell:~\$ (sprint3-analytics-marc-ponce)

Cloud Shell Editor

```

10
11
12
13
14 CREATE OR REPLACE TABLE
15   `sprint3-analytics-marc-ponce.sprint3_silver.users_combined`
16 AS SELECT
17   eu.id AS user_id,
18   eu.name,
19   eu.surname,
20   eu.phone,
21   eu.email,
22   eu.birth_date,
23   eu.country,
24   eu.city,
25   eu.postal_code,
26   eu.address,
27   'EU' AS origin
28 FROM `sprint3-analytics-marc-ponce.sprint3_bronze.european_users_raw` AS eu
29 UNION ALL
30 SELECT
31   au.id AS user_id,
32   au.name,
33   au.surname,
34   au.phone,
35   au.email,
36   au.birth_date,
37   au.country,
38   au.city,
39   au.postal_code,
40   au.address,
41   'US' AS origin
42 FROM `sprint3-analytics-marc-ponce.sprint3_bronze.american_users_raw` AS au;

```

Job Information Results Execution details

This statement replaced the table named users_combined.

Cloud Shell Editor

```

CREATE OR REPLACE TABLE
`sprint3-analytics-marc-ponce.sprint3_silver.users_combined`
AS SELECT
  eu.id AS user_id,
  eu.name,
  eu.surname,
  eu.phone,
  eu.email,
  eu.birth_date,
  eu.country,
  eu.city,
  eu.postal_code,

```

Job Information

Job ID	sprint3-analytics-marc-ponce-europe-southwest1.job_IF3ukM9uKlbZVh4T1AOqDefmaXMy
User	MarcPonceSodhnov@gmail.com
Location	europe-southwest1
Creation time	Sep 29, 2026, 10:52:33 AM
Start time	Sep 29, 2026, 10:52:33 AM
End time	Sep 29, 2026, 10:52:35 AM
Duration	2.119 sec
Bytes processed	581675
Bytes billed	20971520
Slot milliseconds	1048
Job priority	INTERACTIVE
Use legacy SQL	false
Destination table	sprint3-analytics-marc-ponce:sprint3_silver:users_combined
Query	/* @datacloud.settings { "version": 1, "service": "BIG..."
Labels	

Google Cloud | sprint3-analytics-marc-ponce

Search (f) for resources, docs, products, and more

Sandbox Set up billing to upgrade to the full BigQuery experience. [Learn more](#)

Dismiss Upgrade

Explorer + Add data

Search for resources

sprint3-analytics-marc-ponce

Datasets

Connections

Conversations

Agents

Queries

(Classic) Queries

Notebooks

Data canvases

Data preparations

Pipelines

sprint3-analytics-marc-ponce / Datasets / sprint3_silver / Tables / users_combined

users_combined Query Chat Open in - Share Copy Snapshot Delete Export / sync Refresh

Schema Details Preview Insights Lineage Data Profile Data Quality

Filter Enter property name or value

Field name	Type	Mode	Description	Key	Collation	Default Value	Data Governance Tags	Policy Tags	Data Policies
user_id	INTEGER	NULLABLE	-	-	-	-	-	-	-
name	STRING	NULLABLE	-	-	-	-	-	-	-
surname	STRING	NULLABLE	-	-	-	-	-	-	-
phone	STRING	NULLABLE	-	-	-	-	-	-	-
email	STRING	NULLABLE	-	-	-	-	-	-	-
birth_date	STRING	NULLABLE	-	-	-	-	-	-	-
country	STRING	NULLABLE	-	-	-	-	-	-	-
city	STRING	NULLABLE	-	-	-	-	-	-	-
postal_code	STRING	NULLABLE	-	-	-	-	-	-	-
address	STRING	NULLABLE	-	-	-	-	-	-	-
origin	STRING	NULLABLE	-	-	-	-	-	-	-

Edit schema View row access policies Describe data

Exercici 4: Materialització de Companyies i Targetes de Crèdit

Escenari:

Per completar el model de dades a la capa Silver, ens falten dues dimensions clau que actualment depenen de fitxers CSV externs: **Companyies** i **Targetes de Crèdit**. No podem dependre de fitxers solts per a l'anàlisi final. Hem d'importar aquestes dades a taules natives (Silver) i aprofitar per corregir formats.

Tasques:

1. Crea la taula `sprint3_silver.companies_clean`. Copia les dades tal qual, però assegura't que sigui una taula nativa de BigQuery (no una vista externa).

2. Crea la taula `sprint3_silver.credit_cards_clean`. Copia les dades tal qual, però assegura't que sigui una taula nativa de BigQuery (no una vista externa).

Si cal, reanomenar l' `id` segons correspongui.

```
CREATE OR REPLACE TABLE
`sprint3-analytics-marc-ponce.sprint3_silver.companies_clean`
AS SELECT *
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.companies_raw` AS c;

CREATE OR REPLACE TABLE
`sprint3-analytics-marc-ponce.sprint3_silver.credit_cards_clean`
AS SELECT *
FROM `sprint3-analytics-marc-ponce.sprint3_bronze.credit_cards_raw` AS c;
```

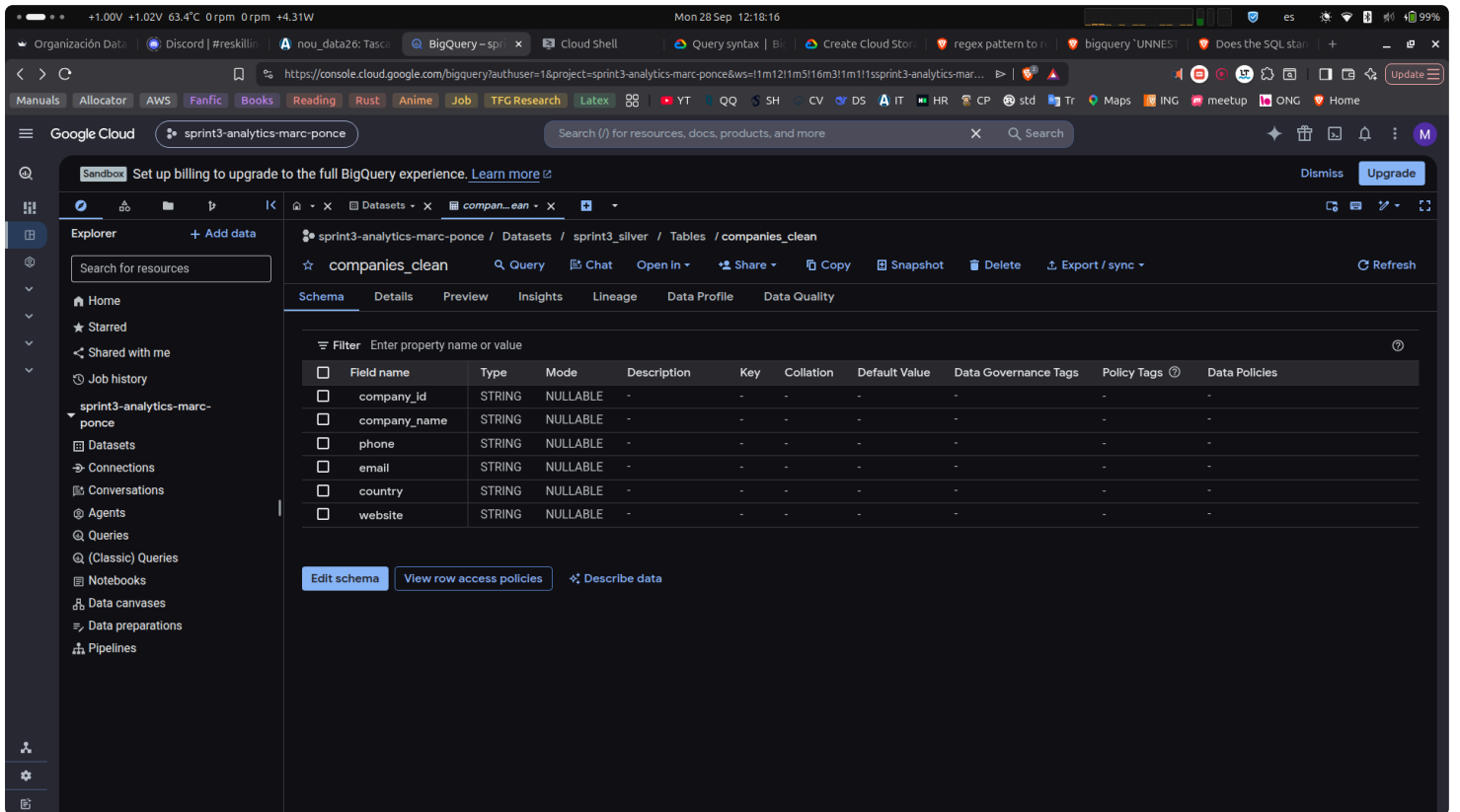
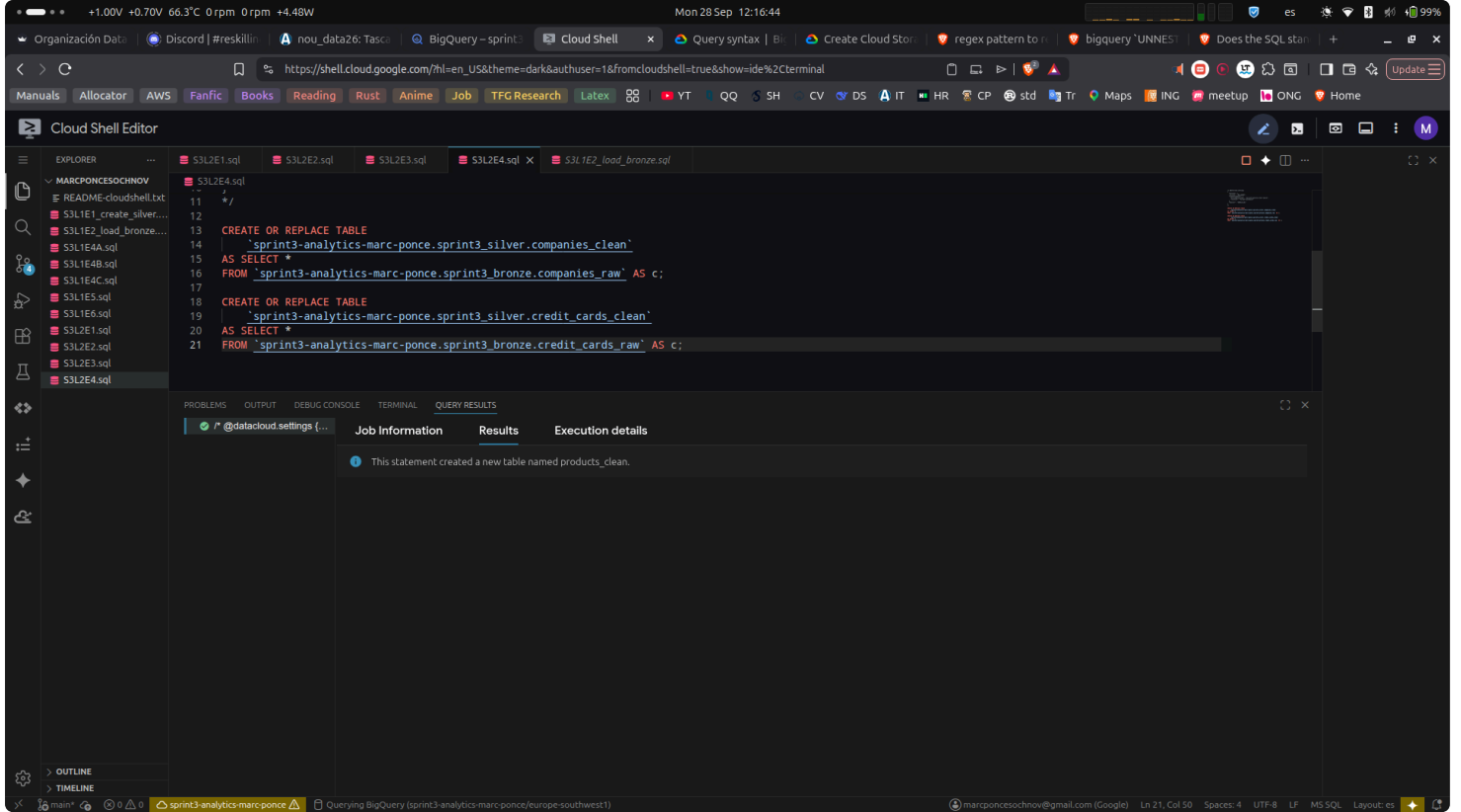
The screenshot shows a Cloud Shell terminal window with the following content:

```
Cloud Shell Editor
EXPLORER
MARCPONCESOCHNOV
  README-cloudshell.txt
  S3L1E1_create_silver...
  S3L1E2_load_bronze...
  S3L1E4A.sql
  S3L1E4B.sql
  S3L1E4C.sql
  S3L1E5.sql
  S3L1E6.sql
  S3L2E1.sql
  S3L2E2.sql
  S3L2E3.sql
  S3L2E4.sql

S3L2E4.sql
1  /* @datacloud.settings
2  {
3    "version": 1,
4    "service": "BIG_QUERY",
5    "connectionInfo": {
6      "billingProjectId": "sprint3-analytics-marc-ponce",
7      "location": "europe-southwest1"
8    },
9    "dialect": "GOOGLE_SQL"
10 }
11 */
12
13 CREATE OR REPLACE TABLE
14 `sprint3-analytics-marc-ponce.sprint3_silver.companies_clean`
15 AS SELECT *
16 FROM `sprint3-analytics-marc-ponce.sprint3_bronze.companies_raw` AS c;
17
18 CREATE OR REPLACE TABLE
19 `sprint3-analytics-marc-ponce.sprint3_silver.credit_cards_clean`
20 AS SELECT *
21 FROM `sprint3-analytics-marc-ponce.sprint3_bronze.credit_cards_raw` AS c;
```

Terminal output:

```
marcpocesochnov@cloudshell:~$ bq query --location=europe-southwest1 --dry run < S3L2E4.sql
Query successfully validated. Assuming the tables are not modified, running this query will process lower bound of 0 bytes of data.
marcpocesochnov@cloudshell:~$
```



The screenshot shows the Google Cloud BigQuery console interface. The left sidebar contains navigation options like Explorer, Home, Starred, Shared with me, Job history, and various data management tools. The main area displays the schema for the 'credit_cards_clean' table within the 'sprint3-analytics-marc-ponce' project. The schema table lists the following columns:

Field name	Type	Mode	Description	Key	Collation	Default Value	Data Governance Tags	Policy Tags	Data Policies
id	STRING	NULLABLE	-	-	-	-	-	-	-
user_id	INTEGER	NULLABLE	-	-	-	-	-	-	-
iban	STRING	NULLABLE	-	-	-	-	-	-	-
pan	STRING	NULLABLE	-	-	-	-	-	-	-
pin	INTEGER	NULLABLE	-	-	-	-	-	-	-
cvv	INTEGER	NULLABLE	-	-	-	-	-	-	-
track1	STRING	NULLABLE	-	-	-	-	-	-	-
track2	STRING	NULLABLE	-	-	-	-	-	-	-
expiring_date	DATE	NULLABLE	-	-	-	-	-	-	-

Below the schema table, there are buttons for 'Edit schema', 'View row access policies', and 'Describe data'.

Nivell 3: Presentació de Dades i Creació de Vistes

Escenari:

Al nivell anterior vas actuar com a Data Engineer netejant les dades (*clean*). Ara toca posar-se el barret d'Analista de Negoci. El problema actual és que les taules estan separades i algunes tenen formats difícils d'explotar (com la llista de productes comprimida en una sola cel·la). En aquest nivell crearàs **Vistes** per facilitar l'accés a la informació i generaràs informes finals.

Objectius:

- Crear Vistes (`CREATE VIEW`) per encapsular lògica de negoci complexa.
- Resoldre el repte de dades agrupades en strings (`SPLIT` + `UNNEST`).
- Exportar resultats per a tercers (`Reverse ETL`).

Exercici 1: La Vista de Màrqueting (Lògica de Negoci)

Màrqueting necessita segmentar els clients corporatius, però no saben fer JOINS. La teva tasca és deixar-los la informació preparada.

Tasca:

Crea una vista anomenada `sprint3_gold.v_marketing_kpis` que mostri la següent informació per a cada companyia:

- **Nom de la companyia, Telèfon i País** (origen: `companies_clean`).
- **Mitjana de compra** (`AVG(amount)` de `transactions_clean`).
- **Classificació de Client (Lògica)**:
- Crea una columna calculada anomenada `client_tier` .
- Si la mitjana de compra és superior a **260€**, etiqueta com a **"Premium"**.
- Si és igual o inferior, etiqueta com a **"Standard"**.

👉 Entrega:

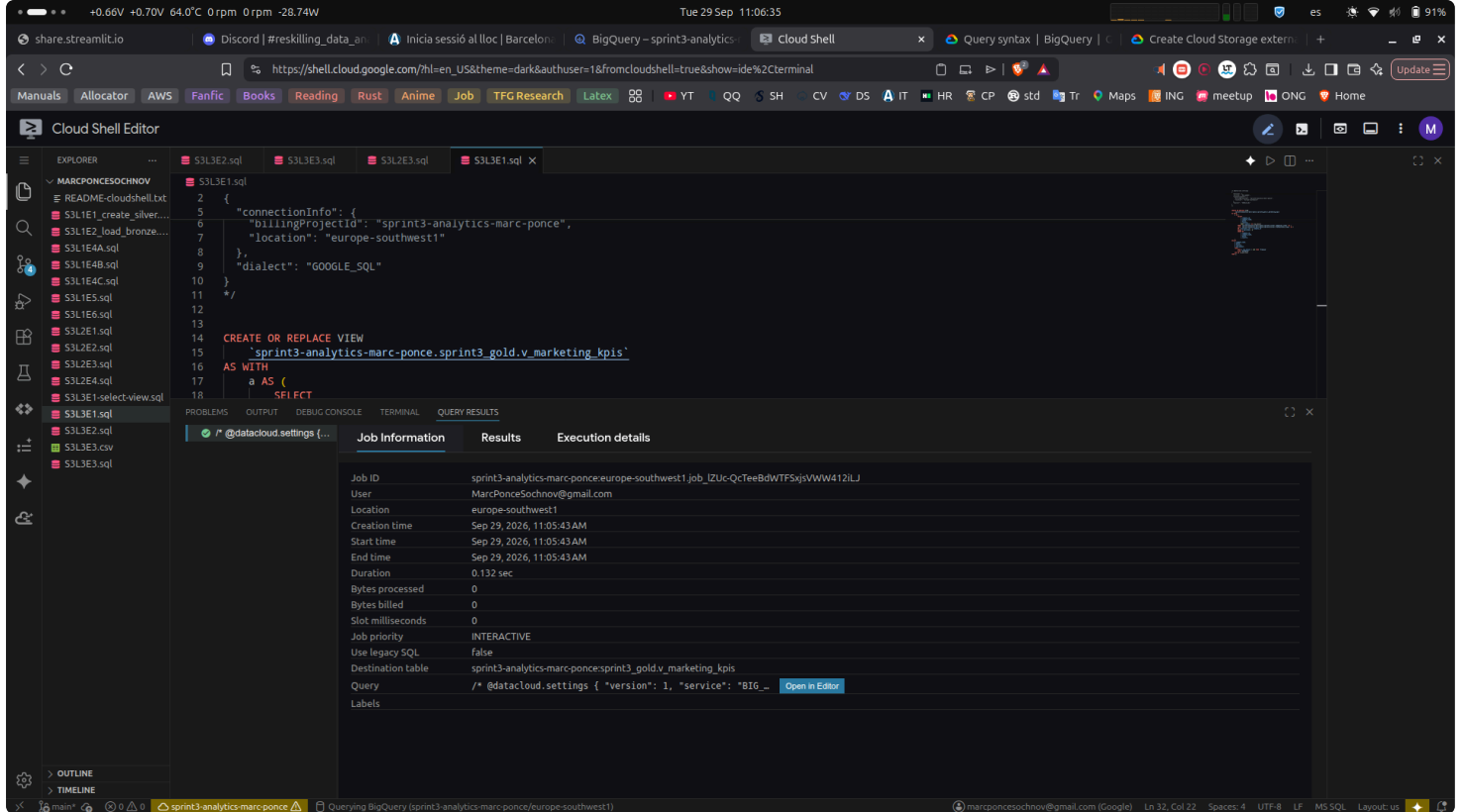
Realitza una consulta `SELECT *` sobre la teva nova vista, ordenant els resultats perquè apareguin primer els clients "Premium" i, dins d'aquests, els que tinguin major mitjana de compra.

```
CREATE OR REPLACE VIEW
`sprint3-analytics-marc-ponce.sprint3_gold.v_marketing_kpis`
AS WITH
a AS (
    SELECT
        c.company_id,
        c.company_name,
        c.phone,
        c.country,
        AVG(t.amount) AS avg_amount
    FROM `sprint3-analytics-marc-ponce.sprint3_silver.companies_clean` AS c
    JOIN `sprint3-analytics-marc-ponce.sprint3_silver.transactions_clean` AS t
    ON t.business_id = c.company_id
    WHERE t.declined = 0
    GROUP BY
        c.company_id,
        c.company_name,
        c.phone,
        c.country
)
SELECT
    a.company_name,
    a.phone,
    a.country,
    a.avg_amount,
    CASE
        WHEN a.avg_amount > 260 THEN 'Premium'
        ELSE 'Standard'
    END AS client_tier
FROM a;
```

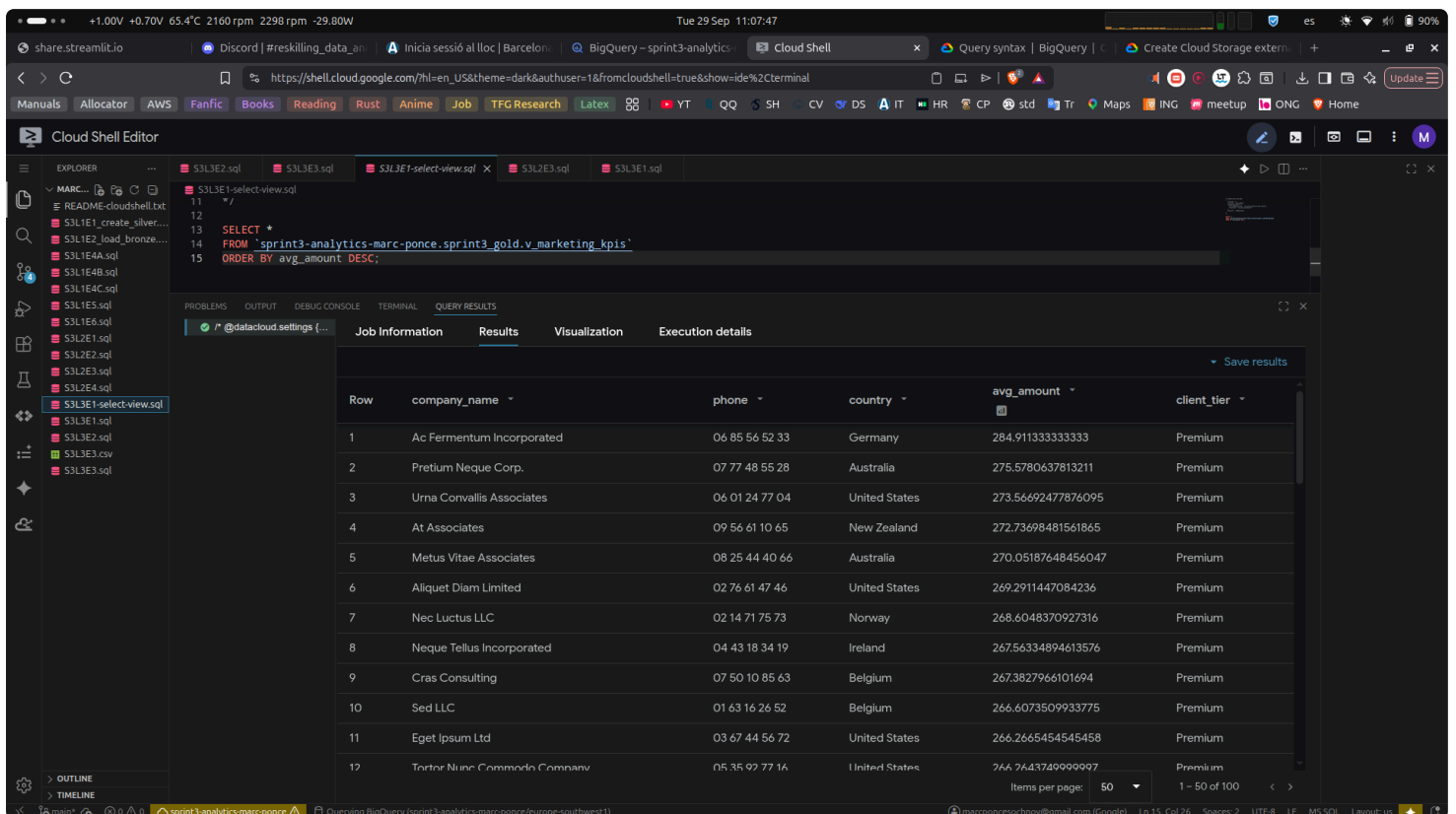
The screenshot shows a Google Cloud Shell terminal window. The top bar displays the URL `https://shell.cloud.google.com/?hl=en_US&theme=dark&authuser=1&fromcloudshell=true&show-ide%2Cterminal`. The left sidebar contains an Explorer panel with a file tree for 'MARCPONCESOCHNOV' and a list of SQL files (S3L1E1.sql to S3L3E3.sql). The main editor displays the content of S3L3E1.sql, which includes a connectionInfo block and a SQL query to create or replace a view named 'sprint3-analytics-marc-ponce.sprint3_gold.v_marketing_kpis'. The query uses a CTE 'a' to select company information and average transaction amounts from 'sprint3-silver.companies_clean' and 'sprint3-silver.transactions_clean' tables, filtered by 't.declined = 0'. The terminal output shows the command `(sprint3-analytics-marc-ponce)? bq query --dry-run < S3L3E1.sql` being executed, followed by a success message: 'Query successfully validated. Assuming the tables are not modified, running this query will process 0 bytes of data.'

This screenshot shows the same Google Cloud Shell terminal window, but with the 'QUERY RESULTS' tab selected. The 'Results' sub-tab is active, displaying a message: 'This statement replaced the view named v_marketing_kpis.' The 'Job Information' and 'Execution details' tabs are also visible. The terminal output at the bottom shows the command `(sprint3-analytics-marc-ponce)?` being entered.

» No bytes billed:



```
SELECT *
FROM `sprint3-analytics-marc-ponce.sprint3_gold.v_marketing_kpis`
ORDER BY avg_amount DESC;
```



Exercici 2: Rànquing de Productes (La Potència dels ARRAY s)

Escenari de Negoci:

L'equip de vendes vol optimitzar el catàleg. Necessiten un informe a la capa Gold que mostri el rendiment real de cada producte. Gràcies a la teva bona enginyeria a la capa Silver, la taula

transactions_clean ja té els IDs de producte perfectament organitzats en una llista (ARRAY). Ara toca explotar aquesta estructura.

Objectiu:

Crea la taula sprint3_gold.product_sales_ranking que contingui l'inventari complet de productes i quantes vegades s'ha venut cadascun.

Requisits de l'Informe:

- **Detall del Producte:** Ha d'incloure product_id , name , price i color (venen de la taula products_clean).
- **Mètrica de Negoci:** Una columna nova total_sold que compti quantes vegades apareix aquest producte a les transaccions.
 - **Integritat:** Han d'aparèixer tots els productes, fins i tot els que tenen 0 vendes (potser cal descatalogar-los).

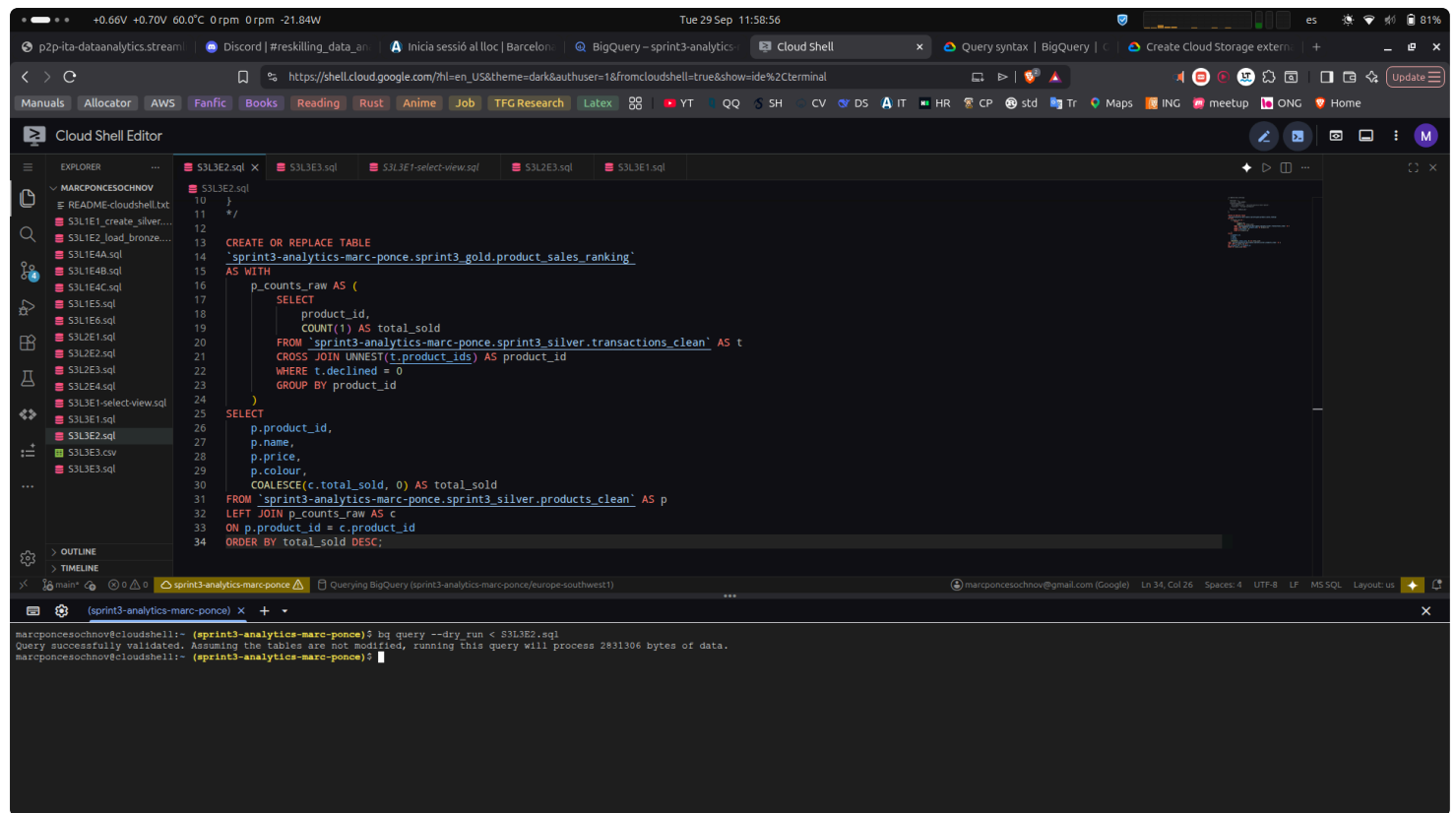
Pista Tècnica:

- A transactions_clean , els productes estan "encapsulats" dins d'un ARRAY per cada transacció.
- Primer necessitaràs "aplanar" la taula de transaccions usant UNNEST(product_ids) per poder comptar els productes individualment.
- Després, fes un encreuament (LEFT JOIN) començant per la taula de productes per no perdre aquells que no s'han venut mai.

```
CREATE OR REPLACE TABLE
`sprint3-analytics-marc-ponce.sprint3_gold.product_sales_ranking`
AS WITH
  p_counts_raw AS (
    SELECT
      product_id,
      COUNT(1) AS total_sold
    FROM `sprint3-analytics-marc-ponce.sprint3_silver.transactions_clean` AS t
    CROSS JOIN UNNEST(t.product_ids) AS product_id
    WHERE t.declined = 0
    GROUP BY product_id
  )
SELECT
  p.product_id,
  p.name,
  p.price,
  p.colour,
  COALESCE(c.total_sold, 0) AS total_sold
FROM `sprint3-analytics-marc-ponce.sprint3_silver.products_clean` AS p
LEFT JOIN p_counts_raw AS c
ON p.product_id = c.product_id
ORDER BY total_sold DESC;
```

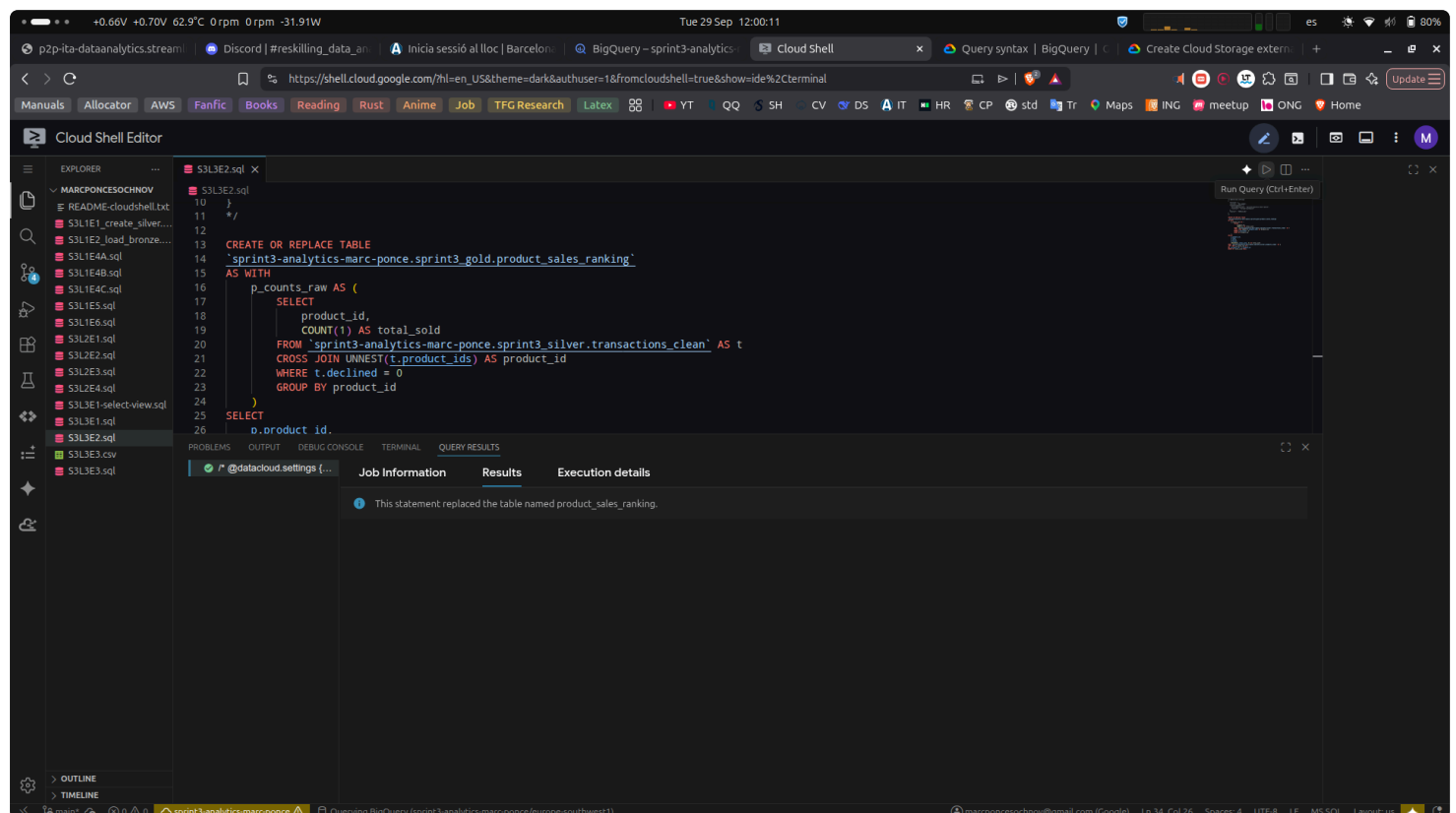
» As it was said earlier UNNEST(...) is used to convert the ARRAY into a virtual table, which we can then JOIN LATERAL with its origin table transactions_clean . BigQuery has LATERAL reserved for

something else, so the syntax is `CROSS JOIN . COALESCE(val0, val1, ...)` yields the first NOT NULL value. In this case it is used so the NULL values left by the `LEFT JOIN` fall back to 0 .



```
10 }
11 */
12
13 CREATE OR REPLACE TABLE
14 `sprint3-analytics-marc-ponce.sprint3_gold.product_sales_ranking`
15 AS WITH
16   p_counts_raw AS (
17     SELECT
18       product_id,
19       COUNT(1) AS total_sold
20     FROM `sprint3-analytics-marc-ponce.sprint3_silver.transactions_clean` AS t
21     CROSS JOIN UNNEST(t.product_ids) AS product_id
22     WHERE t.declined = 0
23     GROUP BY product_id
24   )
25   SELECT
26     p.product_id,
27     p.name,
28     p.price,
29     p.colour,
30     COALESCE(c.total_sold, 0) AS total_sold
31   FROM `sprint3-analytics-marc-ponce.sprint3_silver.products_clean` AS p
32   LEFT JOIN p_counts_raw AS c
33   ON p.product_id = c.product_id
34   ORDER BY total_sold DESC;
```

marcponcesochnov@cloudshell:~\$ bq query --dry_run < S3L3E2.sql
Query successfully validated. Assuming the tables are not modified, running this query will process 283106 bytes of data.
marcponcesochnov@cloudshell:~\$ (sprint3-analytics-marc-ponce)



```
10 }
11 */
12
13 CREATE OR REPLACE TABLE
14 `sprint3-analytics-marc-ponce.sprint3_gold.product_sales_ranking`
15 AS WITH
16   p_counts_raw AS (
17     SELECT
18       product_id,
19       COUNT(1) AS total_sold
20     FROM `sprint3-analytics-marc-ponce.sprint3_silver.transactions_clean` AS t
21     CROSS JOIN UNNEST(t.product_ids) AS product_id
22     WHERE t.declined = 0
23     GROUP BY product_id
24   )
25   SELECT
26     p.product_id,
```

Run Query (Ctrl+Enter)

Job Information Results Execution details

This statement replaced the table named product_sales_ranking.

Cloud Shell Editor interface showing a SQL query in the editor and its execution details in the results pane.

```
CREATE OR REPLACE TABLE
`sprint3-analytics-marc-ponce.sprint3_gold.product_sales_ranking`
AS WITH
  p_counts_raw AS (
    SELECT
      product_id,
      COUNT(1) AS total_sold
    FROM `sprint3-analytics-marc-ponce.sprint3_silver.transactions_clean` AS t
    CROSS JOIN UNNEST(t.product_ids) AS product_id
    WHERE t.declined = 0
    GROUP BY product_id
  )
SELECT
  p.product_id;
```

Job ID	sprint3-analytics-marc-ponce:europa-southwest1 Job_Qe0Lomf0qjEV8zKxkZDsk22TtQ
User	MarcPonceSochnov@gmail.com
Location	europa-southwest1
Creation time	Sep 29, 2026, 12:00:07 PM
Start time	Sep 29, 2026, 12:00:08 PM
End time	Sep 29, 2026, 12:00:08 PM
Duration	0.773 sec
Bytes processed	2831306
Bytes billed	20971520
Slot milliseconds	1849
Job priority	INTERACTIVE
Use legacy SQL	false
Destination table	sprint3-analytics-marc-ponce:sprint3_gold.product_sales_ranking
Query	/* @datacloud.settings { "version": 1, "service": "BIG ...
Labels	

Exercici 3: Exportació de Resultats

El teu màner no té accés a BigQuery i vol el llistat de "Top Productes" en un Excel per a una reunió.

Tasca:

Exporta les dades de la taula `product_sales_ranking` a **Google Sheets** o baixa l'arxiu **CSV** localment.

Entrega:

Una captura de pantalla on es vegi l'arxiu obert (Excel/Sheets) amb les dades correctament formatades.

```
SELECT *
FROM `sprint3-analytics-marc-ponce.sprint3_gold.product_sales_ranking`;
```

Mon 28 Sep 13:54:50

Organización Data Analy | Discord | #reskilling_data | nou_data26: Tasca S2.01 | BigQuery - sprint3-analy | cloud shell

https://shell.cloud.google.com/console#_=_US&theme=dark&authuser=1&fromcloudshell=true&show=ide%2Cterminal

Manuales | Allocator | AWS | Fanfic | Books | Reading | Rust | Anime | Job | TFG Research | Latex | YT | QQ | SH | CV | DS | IT | HR | CP | std | Tr | Maps | ING | meetup | ONG | Home

Cloud Shell Editor

EXPLORER

- MARCPONCESOCHNOV
 - README-cloudshell.txt
 - S3L1E1_create_silver...
 - S3L1E2_load_bronze...
 - S3L1E4A.sql
 - S3L1E4B.sql
 - S3L1E4C.sql
 - S3L1E5.sql
 - S3L1E6.sql
 - S3L2E1.sql
 - S3L2E2.sql
 - S3L2E3.sql
 - S3L2E4.sql
 - S3L3E1-select-view.sql
 - S3L3E1.sql
 - S3L3E2.sql
 - S3L3E3.sql

S3L3E3.sql

```

1 {
2   "version": 1,
3   "service": "BIG_QUERY",
4   "connectionInfo": {
5     "billingProjectId": "sprint3-analytics-marc-ponce",
6     "location": "europe-southwest1"
7   },
8   "dialect": "GOOGLE_SQL"
9 }
10 }
11 */
12
13 SELECT *
14 FROM `sprint3-analytics-marc-ponce.sprint3_gold.product_sales_ranking`

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL QUERY RESULTS

Job Information Results Visualization Execution details

Save results

Row	product_id	name	price	colour	total_sold
1	52	riverlands the duel	31.52	#727272	2654
2	29	Tully maester Tarly	167.2	#111111	2635
3	21	duel Direwolf	96.9	#e2e2e2	2609
4	16	the duel warden	180.91	#666666	2608
5	66	mustafar jinn	2.12	#545454	2601
6	87	sith Jade	96.26	#e2e2e2	2598
7	48	rock Renly in	21.53	#9e9e9e	2597
8	99	the duel warden	127.09	#101010	2597

Items per page: 50 1 - 50 of 100

main* | sprint3-analytics-marc-ponce | Querying BigQuery (sprint3-analytics-marc-ponce/europe-southwest1) | marcpocesochnov@gmail.com (Google) | Ln 14, Col 72 | Spaces: 4 | UTF-8 | LF | MS SQL | Layout: es

```

bq query \
--format=csv \
--max_rows=1000000 \
< S3L3E3.sql \
> S3L3E3.csv

```

Tue 29 Sep 10:28:27

share.streamlit.io | Discord | #reskilling_data | Inicia sessió al llor | Barce | BigQuery - sprint3-analy | cloud shell

https://shell.cloud.google.com/console#_=_US&theme=dark&authuser=1&fromcloudshell=true&show=ide%2Cterminal

Manuales | Allocator | AWS | Fanfic | Books | Reading | Rust | Anime | Job | TFG Research | Latex | YT | QQ | SH | CV | DS | IT | HR | CP | std | Tr | Maps | ING | meetup | ONG | Home

Cloud Shell Editor

EXPLORER

- MARCPONCESOCHNOV
 - README-cloudshell.txt
 - S3L1E1_create_silver...
 - S3L1E2_load_bronze...
 - S3L1E4A.sql
 - S3L1E4B.sql
 - S3L1E4C.sql
 - S3L1E5.sql
 - S3L1E6.sql
 - S3L2E1.sql
 - S3L2E2.sql
 - S3L2E3.sql
 - S3L2E4.sql
 - S3L3E1-select-view.sql
 - S3L3E1.sql
 - S3L3E2.sql
 - S3L3E3.csv
 - S3L3E3.sql

S3L3E3.csv

```

1 product_id,name,price,colour,total_sold
2 52,riverlands the duel,31.52,#727272,2654
3 29,Tully maester Tarly,167.2,#111111,2635
4 21,duel Direwolf,96.9,#e2e2e2,2609
5 16,the duel warden,180.91,#666666,2608
6 66,mustafar jinn,2.12,#545454,2601
7 87,sith Jade,96.26,#e2e2e2,2598
8 48,rock Renly in,21.53,#9e9e9e,2597
9 33,duel warden,127.09,#191919,2597
10 23,riverlands north,169.96,#545454,2593
11 68,Stark Karstark,167.37,#333333,2589
12 88,Stannis warden south,167.15,#515151,2587
13 28,cheWBacca mustafar,127.44,#efefef,2584
14 4,warden south duel,171.89,#111111,2584
15 73,Dorne bastard,114.09,#848484,2584
16 31,Lannister,85.02,#f3f3f3,2580
17 34,of north,120.48,#919191,2580
18 38,ewok sith,55.43,#666666,2578
19 36,Lannister,96.22,#a5a5a5,2575
20 61,Winterfell Lannister,28.01,#848484,2575
21 54,dooku solo,13.14,#e5e5e5,2575

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL QUERY RESULTS

main* | sprint3-analytics-marc-ponce | marcpocesochnov@gmail.com (Google) | Ln 102, Col 1 | Spaces: 4 | UTF-8 | LF | Plain Text | Layout: es

```

marcpocesochnov@cloudshell:~$ (sprint3-analytics-marc-ponce)$ bq query \
--format=csv \
--max_rows=1000000 \
< S3L3E3.sql \
> S3L3E3.csv
marcpocesochnov@cloudshell:~$ (sprint3-analytics-marc-ponce)$

```

S3L3E3																		
File Edit View Insert Format Data Tools Extensions Help																		
Q Menus 100% 123 Defa... 10 B I A																		
A1	product_id																	
1	product_id	name	price	colour	total_sold													
2	52	riverlands the du	31.92	#727272	2654													
3	29	Tully maester Ta	167.2	#111111	2635													
4	21	duel Direwolf	96.9	#e2e2e2	2609													
5	16	the duel warden	180.91	#666666	2608													
6	66	mustafar jinn	2.12	#545454	2601													
7	87	sith Jade	96.26	#e2e2e2	2598													
8	48	rock Renly in	21.53	#9e9e9e	2597													
9	33	duel warden	127.09	#191919	2597													
10	23	riverlands north	169.96	#545454	2593													
11	68	Stark Karstark	167.37	#333333	2589													
12	88	Stannis warden	167.15	#515151	2587													
13	28	chewbacca musi	127.44	#efefef	2584													
14	4	warden south du	71.89	#111111	2584													
15	73	Dorne bastard	114.09	#848484	2584													
16	31	Lannister	85.02	#333333	2580													
17	34	of north	128.48	#919191	2580													
18	38	ewok sith	55.43	#666666	2578													
19	36	Lannister	96.22	#a5a5a5	2575													
20	61	Winterfell Lannis	28.01	#848484	2575													
21	54	dooku solo	13.14	#e5e5e5	2575													
22	24	south duel tourn	48.99	#aaaaaa	2574													
23	11	Karstark Dorne	49.7	#141414	2573													
24	2	Tarly Stark	9.24	#919191	2573													
25	72	north	161.6	#474747	2573													
26	47	Tully	82.15	#919191	2571													
27	10	Karstark Dorne	119.52	#4f4f4f	2571													
28	39	warden	196.65	#a0a0a0	2571													
29	74	duel warden	158.29	#707070	2569													
30	64	Direwolf	195.62	#7b7b7b	2568													
31	65	riverlands the	121.91	#6b6b6b	2567													
32	56	solo palpatine ct	130.59	#2b2b2b	2567													
33	53	kingsblood Little	137.81	#3a3a3a	2564													
34	7	north of Casterly	63.33	#7b7b7b	2563													

Working...